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Module 10:

Operational amplifier

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1 Introduction, structure and function of operational amplifiers

The previous module explained how single-stage and multi-stage transistor amplifiers can be used to amplify signals with low input amplitudes. These amplifier circuits were primarily used in measurement and control technology and, until the 1950s, were constructed discretely from electron tubes or transistors. The development of integrated circuits enabled the miniaturisation of circuits from the end of the 1960s onwards, thereby providing modular components for hardware development. This also applied to the differential amplifiers developed in the 1940s, which, due to their initially widespread use in analogue computers, are also referred to as operational amplifiers (from 'operator'). This module covers the basics of operational amplifiers.

The following skills are to be acquired in this chapter:

Learning objectives: Operationsverstärker

The students can

- describe the structure of a simple operational amplifier.
- explain the operating principle of an operational amplifier.

1.1 Structure and functionality

Operational amplifiers (also referred to as OPV or OpAmp for short) are characterised by the fact that they are designed as universal amplifiers and their function is largely determined by external circuitry. This means that the same component can be used to amplify voltages , perform arithmetic operations such as addition, subtraction or integration, and switch signals. Such operational amplifiers are indispensable in measurement and control technology, signal processing and signal shaping, for example. The following shows an operational amplifier constructed from discrete transistors and an integrated circuit. Due to the direct coupling of the amplifier stages, operational amplifiers can amplify DC and AC voltages and are therefore classified as DC voltage amplifiers.

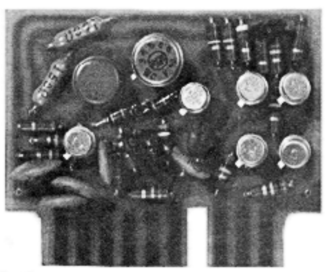


Figure 1.1: Discrete operational amplifier (Quelle: Analog Devices, Wikipedia, Licence CC-BY-SA)

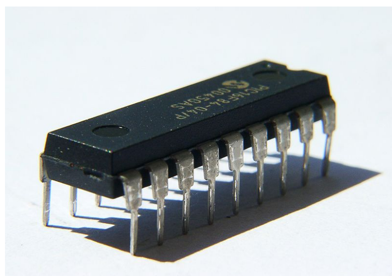


Figure 1.2: Operational amplifier, monolithic, in dual in-line package (DIP) (Quelle: Wollschaf, Wikipedia, (GNU Free Documentation Licence – with attribution))

The mode of operation can be easily explained using a simplified equivalent circuit diagram (see Figure 1.3). This figure shows a simplified equivalent circuit diagram of an OPV made up of bipolar transistors. Today, operational amplifiers are increasingly constructed from field-effect transistors (FETs).

However, the implementation of amplifier circuits with FETs is very similar to that with bipolar transistors, which is why the mode of operation is shown here using one type of transistor. The NPN transistors T_1 and T_2 are identical and form the input of the operational amplifier. The differential voltage U_{Diff} to be amplified is applied between the input marked '+' and the input marked '-'. All currents and voltages occurring in this module can vary over time and are not designated with $U_{\text{E/A/...}}(t)$ but with $U_{\text{E/A/...}}$ for clarity. The inputs are referred to as the non-inverting input '+' and the inverting input '-'. This differential amplifier at the input of the OPV is outlined in red in Figure 1.3. The transistor T_3 functions as a current source when connected to diodes D_1 and D_2. The two diodes regulate the base voltage at T_3 (this part of the circuit is marked in purple).

When the non-inverted input voltage increases (A), the resistance of transistor T_2 decreases. This leads to a reduction in voltage at its collector and an increase at the emitter (B). Since the base of transistor T_4 is connected to the collector of T_2, this causes a corresponding reduction in voltage at the base of T_4 (C). As a result, the collector-emitter path of the PNP transistor T_4 becomes more high-impedance, so that more current flows through the load resistor R_L (D). This increases the voltage drop across R_L and thus the voltage at the output U_A (the output stage of the amplifier is marked in yellow).

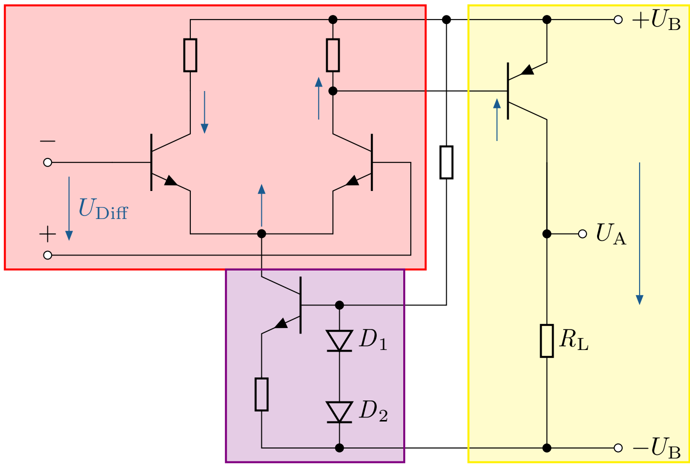


Figure 1.3: Behaviour of a simplified operational amplifier when a positive differential voltage is applied to the input. The blue arrows indicate how the potential at the circuit node shifts relative to a balanced input $U_{\text{Diff}} = 0$.

If, on the other hand, the inverted input voltage increases, the resistance in the collector-emitter path of transistor T_1 decreases (see Figures 1.4). This leads to a decrease in the voltage at the collector of T_1 and a simultaneous increase at the emitter. Since the emitters of T_1 and T_2 are connected to each other, the voltage at the emitter of T_2 also increases. This reduces the voltage difference between the base and emitter of T_2, making its collector-emitter path more high-impedance. As a result, the voltage at the collector of T_2 increases, which increases the base voltage of T_4. This makes transistor T_4 more resistive, which reduces the current flow through resistor R_L and consequently lowers the voltage at R_L. The output voltage of this operational amplifier can thus be controlled between $+U_B$ and $-U_B$.

This circuit already reveals some important characteristics of the real operational amplifier:

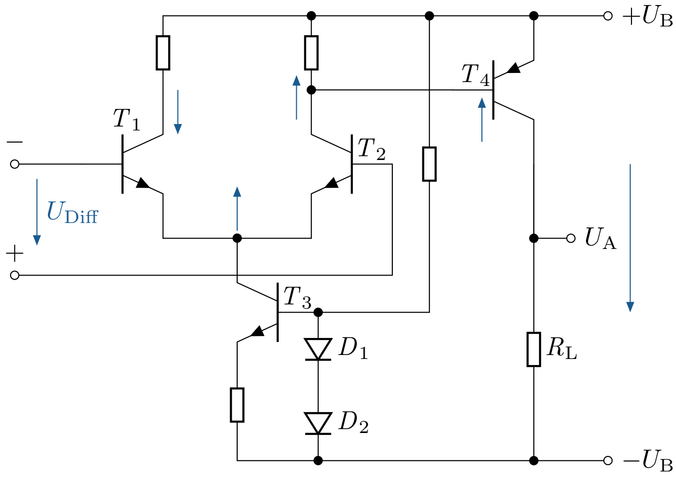


Figure 1.4: Verhalten eines vereinfachten Operationsverstärkers bei Anlegen einer negativen Differenzspannung am Eingang. Die blauen Pfeile geben an, wie sich das Potential am Schaltungsknoten ggü. eines ausgeglichenen Eingangs $U_{Diff} = 0$ verschiebt.

Key point:

1. The input current consumption of the amplifier depends on transistors T_1 and T_2. This current consumption is usually very low, ranging from a few nA to a few hundred pA. The low current consumption is due to the high-impedance collector circuit of the two input transistors.
2. The gain is limited upwards and downwards by the supply voltage.
3. The available output current is determined by the operating voltage source and transistor T_4.

2 Modelling and characteristic variables

The simplified circuit diagram in the previous chapter is well suited for illustrating the internal functioning, but is unsuitable for representation in larger circuits. To make circuit diagrams clearer, operational amplifiers are usually represented by a standard symbol. The internal structure of the amplifier is neglected and characteristic properties are assigned to this general amplifier. Similar to the transistor symbol, the operational amplifier can thus be represented in a simplified form in a circuit.

The following skills are to be acquired within the scope of this chapter:

Learning objectives: Operationsverstärker

The students can

- Describe the differences between the simplified operational amplifier model and real operational amplifiers.
- Specify important areas of the operational amplifier characteristic curve.

The circuit element in Figure 2.1 (left) shows the inverting input, the non-inverting input, the output, the ground reference and the connections for the supply voltages. This circuit symbol is often found in Anglo-American countries and older documents and is only included here for the sake of completeness. The new circuit symbol (right) should be used in this document. If the reference ground is not explicitly shown, this representation assumes that the output is referenced to ground. For didactic reasons, however, the supply voltages are also specified in this document. The ideal amplifier also has the infinity symbol in the upper right corner. This means that the gain is not limited. The advantage of this circuit element is that amplifiers with amplifier limitations can also be modelled. In this case, the symbol for 'infinity' $\widehat{=}\ \infty$ is replaced by a finite value.

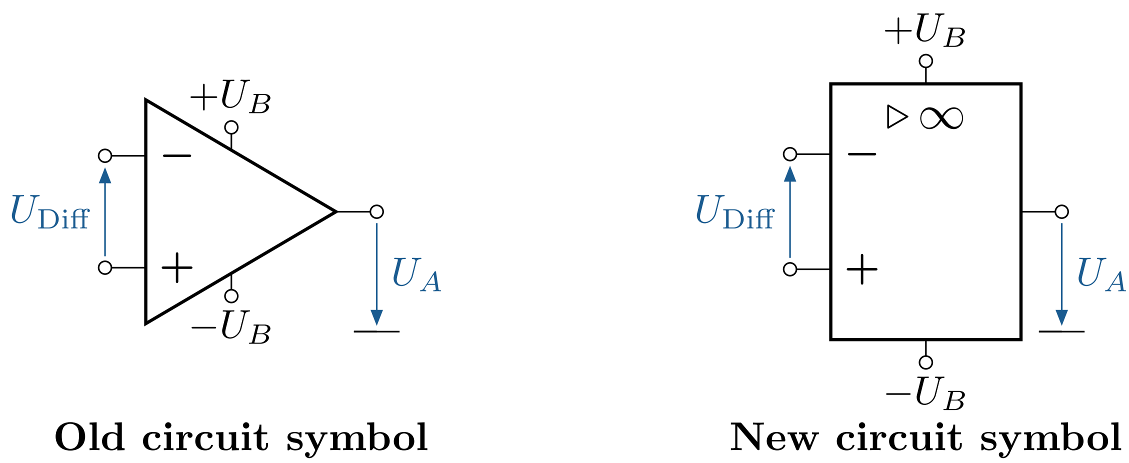


Figure 2.1: Circuit element OPV in old (left) and new according to DIN EN 60617 Part 13 (right)

Two of the four connections[1] of the amplifier, also known as poles, can be combined under idealised conditions to form a so-called „gate“. The operational amplifier can thus be modelled using a so-called two-port model. This is characterised by the fact that, in simplified terms, it can be assumed that the transfer behaviour can be described entirely by the variables current and voltage. The simplification provided by the two-port model means that the internal structure of the operational amplifier can be neglected and regarded as a 'black box'. The previously rather complicated internal wiring of the transistors can be modelled by the component shown in Figure 2.1. In addition, the two-port model assumes that no current flows into the inputs of the operational amplifier [2]. The following table lists additional properties that facilitate calculations with operational amplifiers. The table also shows classic values for a real operational amplifier, which may vary slightly depending on the amplifier type and are only intended to illustrate the limitations of the two-terminal model.

[1] The connections for the supply voltage are not counted here. The output side, on the other hand, is understood as a connection.

[2] This is necessary to satisfy the so-called gate condition. This requires that the current in the inputs must be opposite, i.e. $I_+ = I_-$.

Table 2.1: Ideal and typical operational amplifier characteristics

Designation	**Ideal OPV properties**	**Typical values (e.g. OPA 121)**	**Explanation**
Open-circuit gain	$V_{\text{Leer}} = \infty$	$V_{\text{Leer}} = 10^6$	Amplification of the voltage present between the inputs when the OPV is not connected.
Input impedance	$Z_{\text{i}} = \infty\ \Omega$	$Z_i = 10^{13}\ \Omega$	The load resistance of the OPV as seen from the source.
Output impedance	$Z_{\text{a}} = 0\ \Omega$	$Z_{\text{a}} =$ 50 up to 100 Ω	The resistance at the output of the OPV as seen from a load.
Bandwidth	$B = \infty$ MHz	$B > 2$ MHz	Frequency range in which the amplifier behaviour corresponds to the behaviour specified in the data sheet.
Phase shift	$\varphi = 0^\circ$	$\varphi > 0^\circ$	Delay of the output signal relative to the input signal.
Slew Rate	∞ V	2 mV bis 1000 V/μs	Maximum voltage increase per unit of time.
Output controllability	∞	limited to max. U_{B}	Maximum output voltage value.
Input quiescent current	$I_{\text{Ruhe}} = 0$ pA	$I_{\text{Ruhe}} < 5$ pA	Input current consumption at differential voltage $\Delta U_{\text{Diff}} = 0$.
Input offset current	$I_{\text{Off}} = 0$ pA	$I_{\text{Off}} < I_+ - I_-$	Difference between the two input quiescent currents
Input offset voltage	$U_{\text{Off}} = 0$ mV	$U_{\text{Off}} < 2$ mV	Output DC voltage at $\Delta U_{\text{Diff}} = 0$
Common mode - Rejection	$CMR = \infty$ dB	$CMR = 86$ dB	Indicates how well the amplifier suppresses signals that are equally present at both inputs.

The characteristic curve of an operational amplifier can be constructed using the information provided above. This is shown in the following figure:

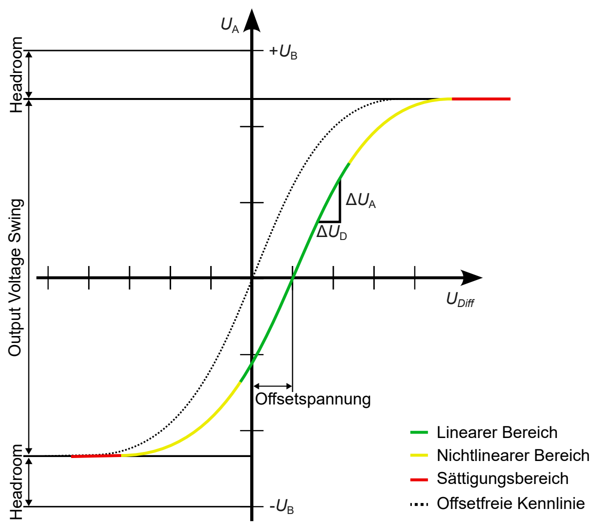


Abbildung 2.2: Kennlinie eines Operationsverstärkers

Figure 2.2 shows the horizontal shift of the amplifier characteristic curve resulting from the offset voltage U_{Off}. It can also be seen that the amplifier exhibits linear amplification behaviour with respect to the applied input voltage within its specified operating range (shown in green). The slope in the linear range depends on the maximum amplification of the operational amplifier used. If this specified operating range is exceeded, the output voltage initially exhibits a non-linear relationship to the input voltage (shown in yellow). However, the output voltage continues to rise until saturation is reached. In the area marked in red, the output voltage remains constant even if the input differential voltage continues to increase. As long as the output voltage is within the operating range of the operational amplifier, a constantly amplified output voltage with an offset is output. The actual achievable output voltage is specified by the variable „Output Voltage Swing“. The difference to the positive or negative supply voltage ($+U_{\mathrm{B}}$ and $-U_{\mathrm{B}}$) is referred to as „Headroom“. The headroom is the result of a PN junction in the semiconductor components used in the operational amplifier and is therefore usually 0.6 - 0.7 V. Headroom is an important parameter for the design of operational amplifiers. Operational amplifiers with very low headroom are usually classified by manufacturers as „rail-to-rail“ amplifiers.

Key point:

To simplify calculations with operational amplifiers, a simplified model is often used. It is important to be aware of the limitations of this model and to take them into account when selecting components and designing circuits.

3 Principle of negative feedback

After the previous chapters initially addressed the internal structure of the amplifier and the resulting model, as well as the real effects not described in the ideal model, this subchapter will deal with the circuitry of the operational amplifier.

The following skills should be acquired within the scope of this chapter:

Learning objectives: Operationsverstärker

The students can

- Name various operational amplifier circuits and possible areas of application.
- Determine the function of existing operational amplifier circuits and calculate the gain using Kirchhoff's laws.
- Name important properties of operational amplifier circuits.

To this end, a problem will be presented for which an operational amplifier is to be used to find a solution. This problem will be used to motivate the use of operational amplifiers step by step.

Example 3.1: Part 1: Development of an amplifier circuit for music signals

When developing hi-fi devices, amplifiers must be matched to the load (i.e., the speaker). For this purpose, a power amplifier (i.e., a final stage) is required. Such an amplifier, capable of boosting the signal from a mobile phone to drive a speaker, is to be built here. A simplified equivalent circuit of the speaker is represented by an inductance in series with a resistor (see Figure 3.1). The input voltage level is 1 V, and a 4 Ω speaker is to be driven (4 Ω means that the speaker has an impedance of 4 Ω at 1 kHz). Determine the required output voltage of the amplifier so that the speaker consumes 25 W at 1 kHz. How can this be approached if an operational amplifier circuit is used for the solution?

Solution

First, the voltage that the amplifier is to output must be calculated.

$$U_\mathrm{A} = \sqrt{P \cdot R} = \sqrt{25\ \mathrm{W} \cdot 4\ \Omega} = 10\ \mathrm{V} \tag{3.1}$$

How can an operational amplifier be used to replace the part of the circuit marked with „?“ in order to achieve the required gain of 10 (see Figure 3.1)?

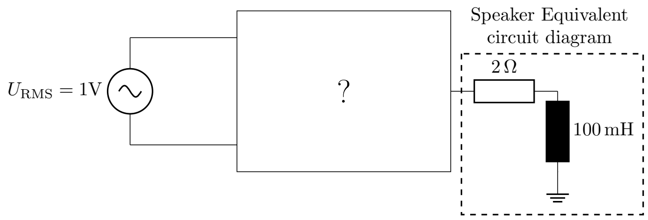


Figure 3.1: Circuit of the example problem with a signal source on the input side and a complex load on the output side. The amplification factor $V = \frac{U_\mathrm{A}}{U_\mathrm{E}}$ is first sought for the part of the circuit marked with a question mark.

As we know from previous chapters, operational amplifiers have an inverting and a non-inverting input. If there is a voltage difference between these two inputs, the operational amplifier amplifies this difference many times over until the voltage reaches the maximum output voltage of the operational amplifier (see A in Figure 3.2). The slew rate indicates how quickly this increase can occur. If the output voltage is not fed back to the input, the output voltage increases until it reaches the level of the supply voltage. This type of amplifier circuit is referred to as a comparator circuit.

This behaviour is undesirable for most applications in which the input voltage signal with low amplitude is to be amplified. Instead, operational amplifiers are usually operated in what is known as negative feedback (B), in which the output is fed back to the inverting input. The switch shown in (B) is initially open and the operational amplifier is not supplied with voltage. At time 0 s, the switch is closed and the supply voltage of the amplifier is switched on, activating the feedback of the output signal U_A (C). The feedback of the output signal ensures that the voltage at the inverting input is adjusted to the voltage level of the output voltage. The amplifier circuit shown in (C) has a gain of 1 and is referred to as an impedance converter, as the amplifier in this type of circuit can be used to implement a circuit with a high-impedance input due to its high input impedance (this is necessary, for example, in measuring circuits for voltage measurements). However, this circuit is not yet suitable for solving the application problem, as a gain of 5 is required. To achieve this, additional resistors are now incorporated into the amplifier circuit (as shown in (D))[3].

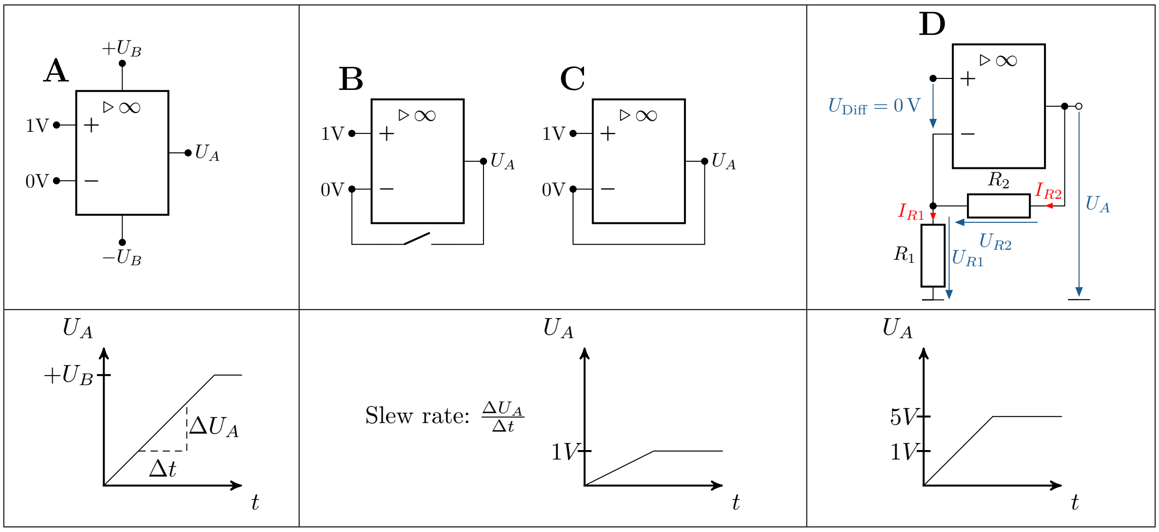


Abbildung 3.2: Circuitry of an amplifier as a comparator (A), impedance converter (C) and non-inverting amplifier (D)

Example 3.2: Part 2: Development of an amplifier circuit for music signals

The final step is to determine how the resistors in (D) must be selected in order to achieve a gain of 5. To do this, the mesh equations must first be established. An important assumption for establishing this mesh equation is that the amplifier equalises the two input signals through the feedback of the output signal. Here, we can write „Assumption: $U_\mathrm{dif} = 0$ V“. This results in $U_\mathrm{E\text{-}} = 1$ V for the inverting input. Since it is further assumed that no current flows into

[3] It can be seen that for $R_2 = 0$ and $R_1 = \infty$, the circuit of the impedance converter is obtained again. The impedance converter is therefore a special case of the non-inverting amplifier.

the inputs, $I_{\mathrm{R1}} = I_{\mathrm{R2}}$. Based on these assumptions, the mesh equation is

$$U_{\mathrm{A}} = U_{\mathrm{R1}} + U_{\mathrm{R2}} = U_{\mathrm{E}} + I_{\mathrm{R2}} \cdot R_2 = U_{\mathrm{E}} + \frac{U_{\mathrm{E}}}{R_1} \cdot R_2 \tag{3.2}$$

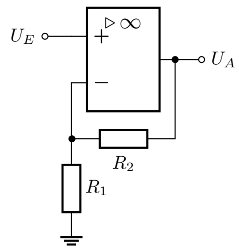
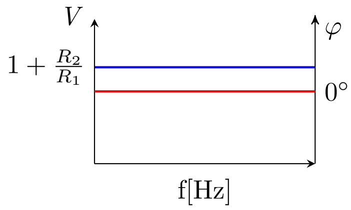
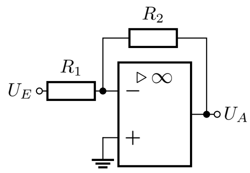
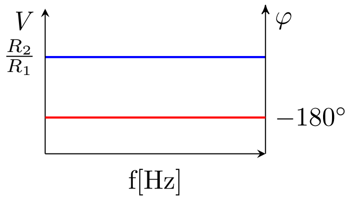
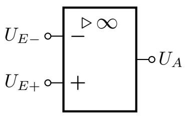
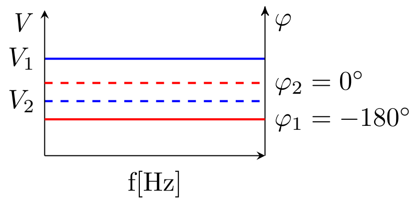
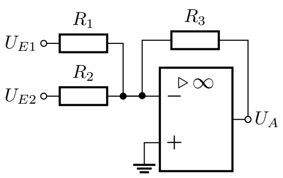
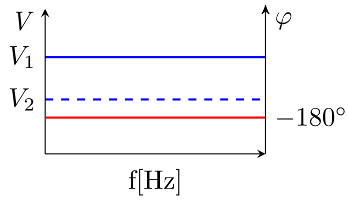
This results in the following expression for the input-output behaviour in the time domain.

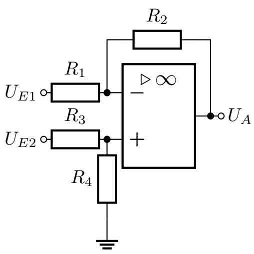
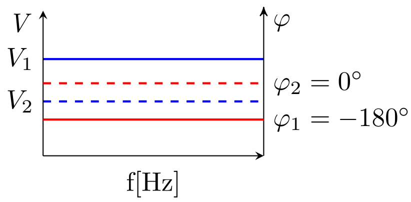
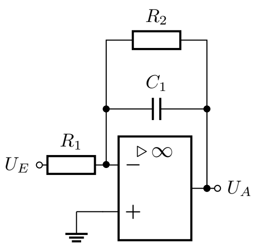
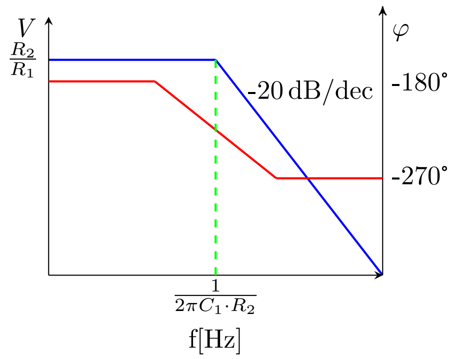
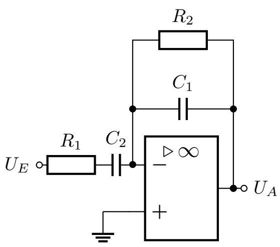
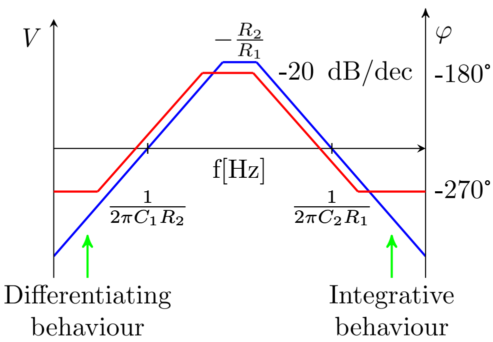
$$\frac{U_{\mathrm{A}}}{U_{\mathrm{E}}} = \underbrace{(1 + \frac{R_2}{R_1})}_{V} \tag{3.3}$$

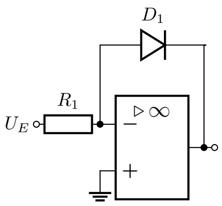
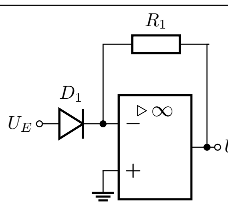
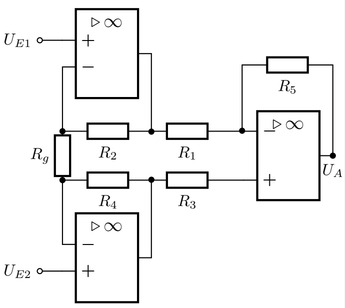
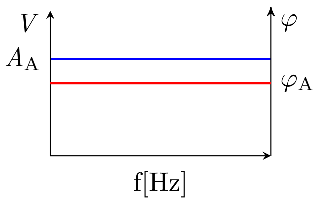
Solving this equation for R_2/R_1 gives the ratio of resistances for the problem described above as $R_2/R_1 = 9$. The resistances should not be chosen too large here, as the balancing current between the output and input could then become too small. This can lead to severe noise at the amplifier output. Typical resistance values can usually be found in the example circuits in the data sheet of an operational amplifier. In this case, for example, $R_2 = 9\ \mathrm{k}\Omega$ and $R_1 = 1\ \mathrm{k}\Omega$ could be selected.

Similar to the circuit shown above, transfer functions can also be derived for all other operational amplifier circuits. However, this will not be shown here. Further transfer functions for some selected basic operational amplifier circuits can be found in Table 3.1. This table shows the circuits, the Bode diagrams with phase response (red) and amplitude response (blue), and the transfer functions. Now that the circuitry of amplifiers has been presented, the following chapter will discuss another important property of amplifier circuits that results from the amplifier components used and the external circuitry: the stability of the amplifier circuit.

Table 3.1: Selected basic operational amplifier circuits

Circuit	Frequency response and equation	Explanation and properties
	 $U_A = \left(1+\frac{R_2}{R_1}\right) U_E$	**Non-inverting amplifier** • Phase-matched output signal • Very high input impedance • Low output impedance • Applications: e.g. impedance converters
	 $U_A = \frac{R_2}{R_1} U_E$	**Inverting amplifier** • -180° phase shift between input and output • R_1 determines the input resistance • Low-impedance output-resistance • Applications: e.g. active voltage divider for measuring high voltages when $R_1 > R_2$
	 $V_1 : U_{E+} > U_{E-} \Rightarrow U_A \approx +U_B$ $V_2 : U_{E+} < U_{E-} \Rightarrow U_A \approx -U_B$	**Comparator** Use in two-point controllers and analogue-to-digital converters
	 $U_A = \underbrace{\frac{R_3}{R_1}}_{V_1} \cdot U_{E1} + \underbrace{\frac{R_3}{R_2}}_{V_2} \cdot U_{E2}$	**Summariser** The summing amplifier is based on the inverting amplifier and is used in analogue computers and for mixing voltage signals.

Circuit	Frequency response and equation	Explanation and properties
	 $U_A = U_{\mathrm{E2}} \cdot \underbrace{\frac{R_1 + R_2}{R_1} \cdot \frac{R_4}{R_3 + R_4}}_{V_2} - U_{\mathrm{E1}} \cdot \underbrace{\frac{R_2}{R_1}}_{V_1}$	**Subtractor** When all resistances are equal, the difference between the signals U_{E1} and U_{E2} is formed, which is why this circuit is referred to as a differential amplifier.
	 $U_A(t) = -\frac{R_2}{R_1} U_{\mathrm{E}}(t) - \int_0^t \frac{U_{\mathrm{E}}(t)}{R_1 C_1}\, dt$	**Integrator** • Integrates the input signal • Used as an active low-pass filter
	 $U_A = -\frac{R_2}{R_1} U_{\mathrm{E}}(t)$ $- \int_0^t \frac{1}{R_1 C_1} U_{\mathrm{E}}(t)\, dt - R_2 C_2 \frac{dU_{\mathrm{E}}(t)}{dt}$	**Differentiator** • Integrates the input signal • Used as an active high-pass filter (in reality, it usually exhibits band-pass behaviour, as shown here)

Circuit	Frequency response and equation	Explanation and properties
	$U_\mathrm{A} = -U_\mathrm{T} \cdot \ln\left(\frac{U_\mathrm{E}}{R_1 \cdot I_S}\right)$ $U_\mathrm{T} = \frac{k_\mathrm{B} \cdot T}{e}$ $e =$ Elementary charge $k_\mathrm{B} =$ Boltzmann constant $I_\mathrm{s} =$ Reverse voltage of the diode	**logarithm calculator** Forms the natural logarithm of the input signal
	$U_\mathrm{A} = -R_1 \cdot I_\mathrm{S} \cdot e^{\frac{U_\mathrm{E}}{U_\mathrm{T}}}$	**Potentiator** Has an e-functional relationship between input and output voltage
	 a_i : Amplitude input voltage i φ_i : Phase input voltage i $A_A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2\cos(\varphi_1 - \varphi_2)}$ $\tan(\varphi_A) = \frac{a_1\sin(\varphi_1) + a_2\sin(\varphi_2)}{a_1\cos(\varphi_1) + a_2\cos(\varphi_2)}$ When : $R_2 = R_4 = R$ $U_\mathrm{A} = (1 + \frac{2R}{R_\mathrm{g}}) \cdot \frac{R_3}{R_2}(U_\mathrm{E2} - U_\mathrm{E1})$	**Instrument amplifier** Differential amplifier with high input impedance and high common-mode rejection

4 Stability of amplifier circuits

Feedback of the output signal to the input signal results in a control loop. However, this control loop is not always stable, i.e. feedback does not mean that the output variable only changes as long as there is a differential voltage at the input[4]. This becomes clear when the output signal is fed back to the non-inverting input instead of the inverting input. If a voltage difference is now applied to the input and amplified at the output, this output voltage leads to a renewed increase in the voltage difference at the input. The system is therefore not stable. A similar phenomenon can be observed when energy storage devices (capacitors or coils) are used in the amplifier circuit. Such a circuit always has a stability limit, which is reached when the signal fed back to the inverting input has a phase shift of 180°. In this case, the negative feedback becomes positive feedback and the circuit becomes unstable. However, this is not the only way in which a circuit can become unstable. The circuit can also start to oscillate if the energy storage devices are not correctly dimensioned. For this reason, stability analyses of the designed circuits are essential.

The following skills are to be acquired in this chapter:

Learning objectives: Operational amplifier

The students can

- Perform stability analyses and evaluate the results of the analysis.
- Dimension components in OPV circuits according to stability criteria.

Some of the stability criteria used for stability assessment are described below. Assessing the stability of operational amplifier circuits is an important skill for electrical engineering students, whether they are designing circuits from scratch or troubleshooting existing circuits. This assessment ensures that operational amplifier circuits work reliably and effectively, which is essential for their integration into various electronic systems. The methods used to assess stability are explained below.

Frequency response analysis (experimental/simulative)

A fundamental technique is frequency response analysis. This method examines how the gain and phase of an operational amplifier circuit change when the frequency of the input signal varies. Tools such as Bode diagrams are often used to visualise this response and help identify potential stability issues such as gain spikes or phase shifts that could lead to instability. An important criterion here is phase margin. Determining this quantity is a quantitative approach to stability assessment. It helps measure the stability margin within a feedback system by considering the phase difference between the actual phase shift at the passband frequency (i.e., 0 dB) and 180°. This point is referred to as the „critical point“ because, with a phase shift greater than 180° and a gain greater than 0 dB, the feedback becomes positive feedback. If the phase difference exceeds a certain threshold value, typically around 45°, it is assumed that the system is far enough away from the stability limit. This analysis is usually performed graphically based on the Bode diagram.

[4] This is a simplification. In control engineering, a distinction is made between Lyapunov stability and BIBO stability (see, for example, „Control Engineering“ by Otto Föllinger)

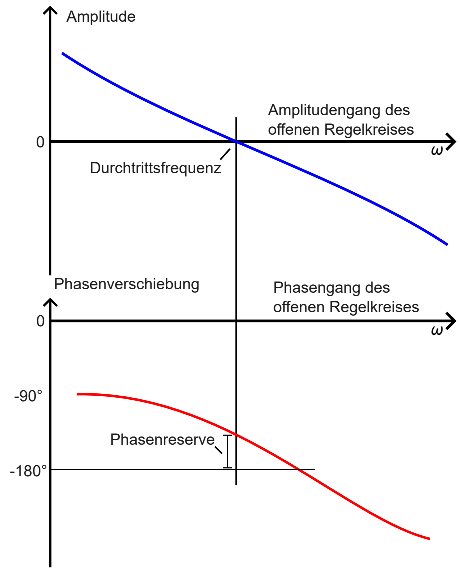


Figure 4.1: Determination of phase reserve based on the Bode diagram

Analysis of the transfer function (analytical)

If the transfer function of the operational amplifier circuit is available, a stability analysis can also be performed directly. To do this, the poles of the transfer function (e.g. in the Laplace domain) are analysed. If these are in the left s-half-plane (i.e. all poles are negative), the circuit has stable transfer behaviour. With this approach, as well as with frequency analysis (unless the Bode diagram has been determined experimentally), it must be noted that the transfer function is usually only valid for a certain frequency range, since, for example, the dynamics of the operational amplifier are not usually taken into account in these methods.

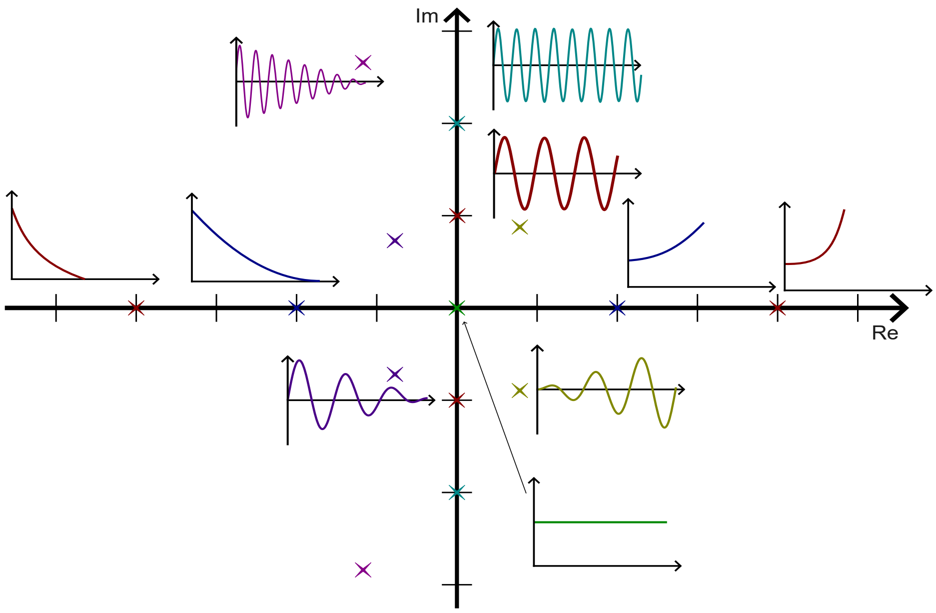


Figure 4.2: System responses at different poll positions

Analysis of transient behaviour (experimental/simulation)

Another important method is the analysis of transient behaviour. In this method, a voltage jump is applied to the amplifier circuit to examine how an operational amplifier circuit reacts to sudden changes in the input signals. By observing the reaction of the circuit, it is possible to determine whether the system is prone to instability. In reality, this test must be carried out very carefully, as components can quickly be destroyed in an unstable circuit. For this reason, such tests are now often carried out purely as simulations. As a rule of thumb, if the circuit tends towards a fixed value after excitation by a unit step and does not oscillate continuously, the transient behaviour can be assumed to be stable.

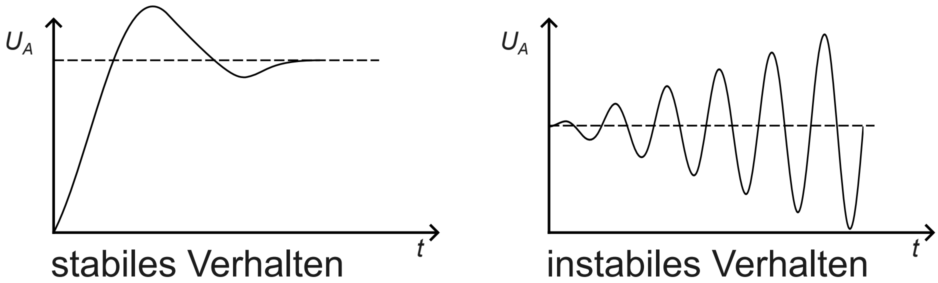


Figure 4.3: Stable and unstable transient behaviour

As explained, stability analysis is a complex topic and will only be touched upon here. No claim is made to completeness at this point. The analysis methods mentioned are part of control engineering and can be found in the relevant literature.

Key point:

Feedback of the output signal to the input results in a control loop. The stability of this control loop must be verified by simulation, experiment or analysis.

5 Operational amplifiers as analogue computers

In addition to their use in measuring amplifiers, OPVs are also used in analogue computers. This may no longer seem relevant in the age of high-performance processors, but it does have advantages. For example, operational amplifiers can significantly reduce calculation time. This is particularly useful in applications where only one arithmetic operation (multiplication, addition, etc.) is to be performed, but where virtually no latency is permitted. This is still often the case in control engineering today.

The following skills are to be acquired within the scope of this chapter:

Learning objectives: Operationsverstärker

The students can

- Specify suitable operational amplifier circuits for solving a problem.
- Calculate resistance ratios.

The following section presents an example that shows how an analogue computer can be constructed using the basic operational amplifier circuits given in the table.

Example 5.1: Beispiel Analogrechner

A circuit is to be designed that implements the following function:

$$U_{\mathrm{A}} = \int U_{\mathrm{E1}} dt + x \cdot U_{\mathrm{E1}} - 2 \cdot U_{\mathrm{E2}} \tag{5.1}$$

Solution
First, the equation must be broken down into two sub-problems that can be solved using operational amplifier circuits.
The first sub-problem is the integration of the input signal U_{E1}. To do this, an integrator circuit should first be used. The second sub-problem is the addition of the signals. A summing amplifier can be used for this. The desired behaviour can therefore be achieved by combining an integrator circuit and a summing amplifier. Is it possible to select x=0 with this circuit?

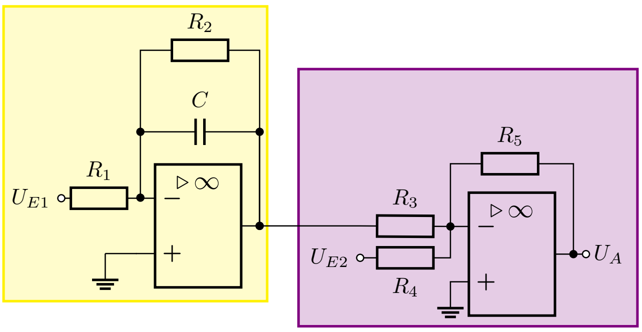


Figure 5.1: Circuit for solving the analogue computer task

$$U_{\mathrm{A}} = \underbrace{\left[-\frac{1}{R_1 \cdot C} \int_0^t U_{\mathrm{E1}}\ dt - \frac{R_2}{R_1} \cdot U_{\mathrm{E1}} \right]}_{Formel\ des\ Integrators} \underbrace{\cdot \left(-\frac{R_5}{R_3} \right) - \frac{R_5}{R_4} \cdot U_{\mathrm{E2}}}_{Formel\ des\ Summierers} \tag{5.2}$$

This can now be transformed as follows

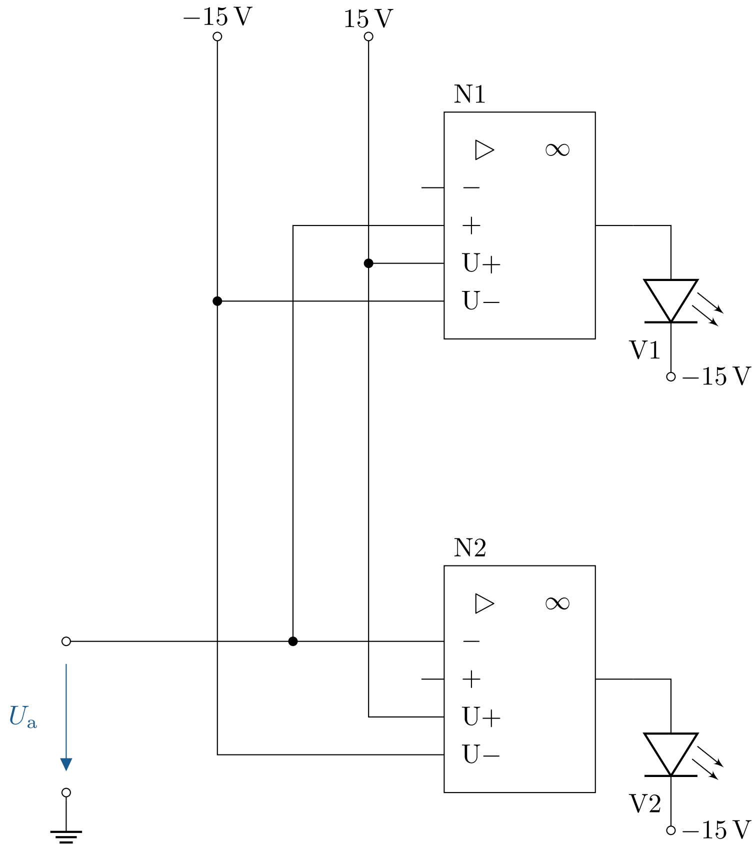
$$U_{\mathrm{A}} = \underbrace{\frac{R_5}{R_1 \cdot R_3 \cdot C}}_{\overset{!}{=}1} \int_0^t U_{\mathrm{E1}}\ dt + \underbrace{\frac{R_2 R_5}{R_1 R_3}}_{\overset{!}{=}x} U_{\mathrm{E1}} - \underbrace{\frac{R_5}{R_4}}_{\overset{!}{=}2} \cdot U_{\mathrm{E2}} \tag{5.3}$$

As can be seen from the formula, the component values now only need to be selected so that the correct pre-factors are obtained. A choice of x=0 is only possible if the resistor R_2 is omitted. In this case, the result is an „ideal“, which, however, is not usually constructed in reality.

A Exercise tasks

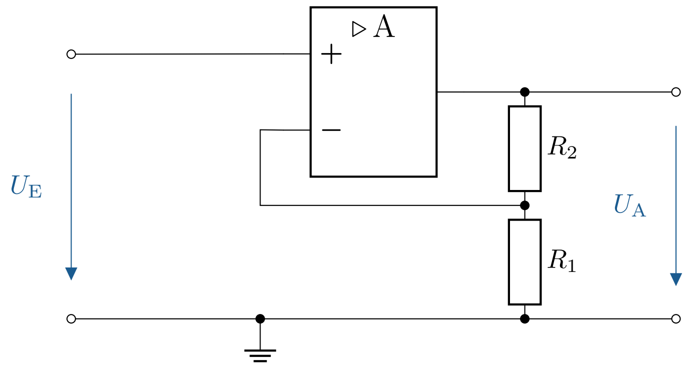
A.1 Monitoring a voltage with LEDs

The circuit shown is given; it is to be used to monitor the voltage U_a. U_a may only lie within the following range: $4.75\,\text{V} < U_a < 5.25\,\text{V}$. If the voltage exceeds or falls below this range, one of the LEDs should light up. Design the circuit.



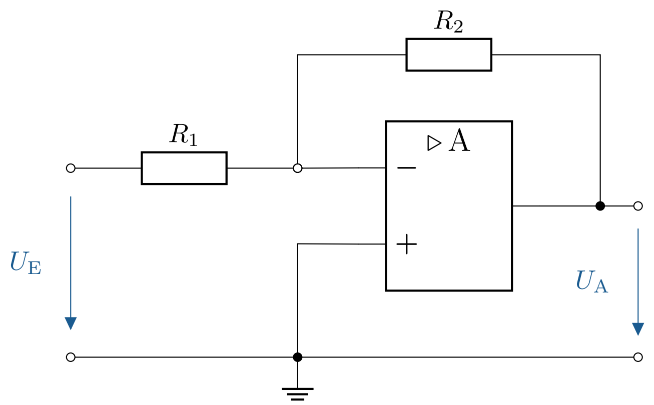
A.2 Resistance calculation for non-inverting amplifiers

The depicted circuit with a non-inverting amplifier is given. The resistor $R_2 = 1\,\text{k}\Omega$. What values must R_1 have so that the gain is 10, 50, and 100?



A.3 Resistance calculation for inverting amplifier

The depicted circuit with an inverting amplifier is given. The resistor $R_2 = 1\,\mathrm{k\Omega}$. What values must R_1 have so that the gain is -10, -50, and -100?

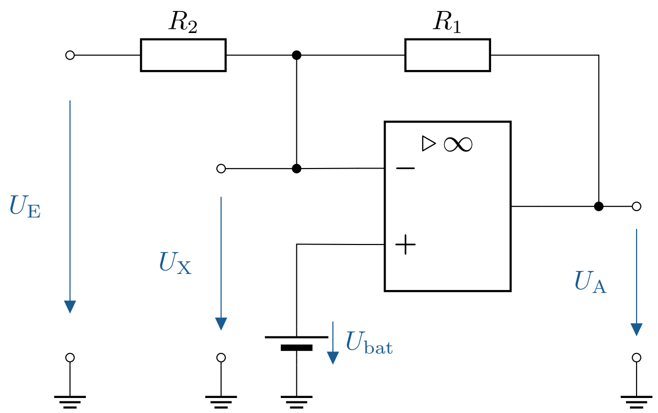


A.4 Calculation of the output voltage of an operational amplifier circuit with two inputs

The depicted circuit with an ideal operational amplifier is given. The resistors are $R_1 = 100\,\mathrm{k\Omega}$ and $R_2 = 10\,\mathrm{k\Omega}$. The supply voltage is $\pm 15\,\mathrm{V}$.

a) Determine the output voltage U_A bei $U_\mathrm{E} = 2\,\mathrm{mV}$ and $U_\mathrm{bat} = 0\,\mathrm{V}$

b) Determine the output voltage U_A bei $U_\mathrm{E} = 20\,\mathrm{mV}$ and $U_\mathrm{bat} = 0\,\mathrm{V}$



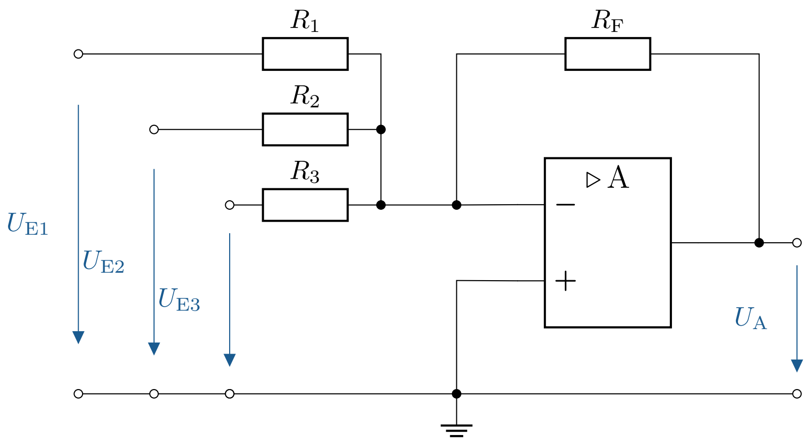
A.5 Calculate the output voltage of an operational amplifier with multiple inputs

The depicted circuit with an ideal operational amplifier is given. The supply voltage is $\pm 15\,\mathrm{V}$.

a) All resistors are equal: $R_1 = R_2 = R_3 = R_\mathrm{F} = 10\,\mathrm{k\Omega}$. The input voltages are $U_\mathrm{E1} = 1\,\mathrm{V}$, $U_\mathrm{E2} = 2\,\mathrm{V}$, $U_\mathrm{E3} = 3\,\mathrm{V}$. What is the output voltage U_A?

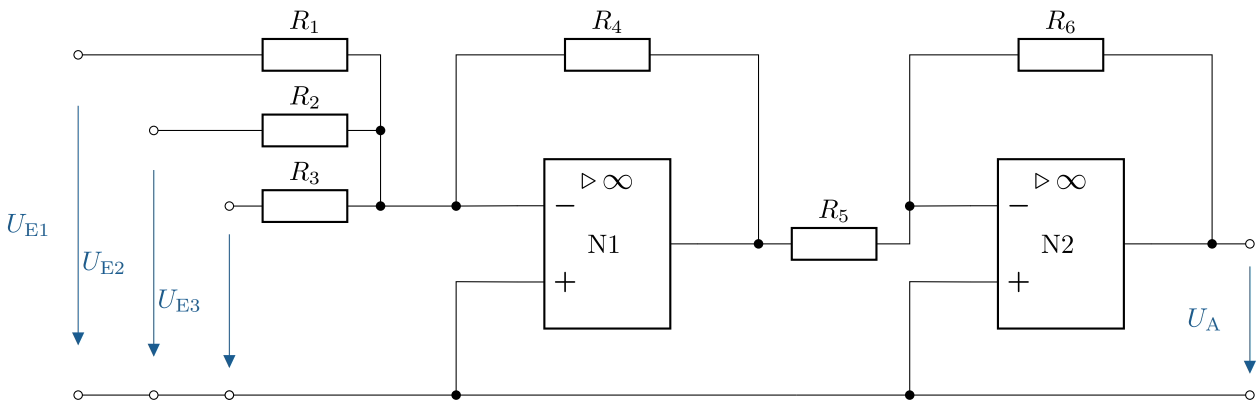
b) The input voltages remain the same, but the resistors are now $R_\mathrm{F} = 10\,\mathrm{k\Omega}$, $R_1 = 2\,\mathrm{k\Omega}$, $R_2 = 5\,\mathrm{k\Omega}$, $R_3 = 10\,\mathrm{k\Omega}$. What is the output voltage U_A?

c) The input voltages remain the same, but the resistors are now $R_F = 10\,\text{k}\Omega$, $R_1 = 10\,\text{k}\Omega$, $R_2 = 5\,\text{k}\Omega$, $R_3 = 2\,\text{k}\Omega$. What is the output voltage U_A?



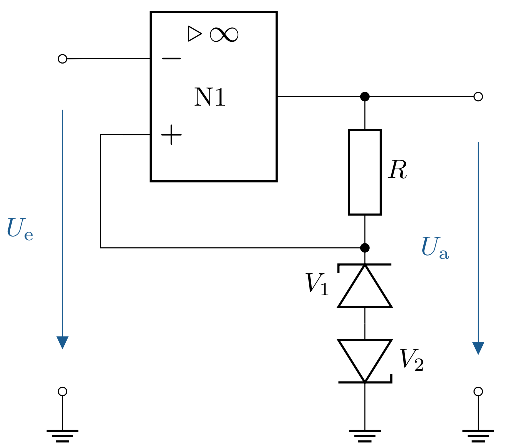
A.6 Calculation of the output voltage of a two-stage operational amplifier circuit

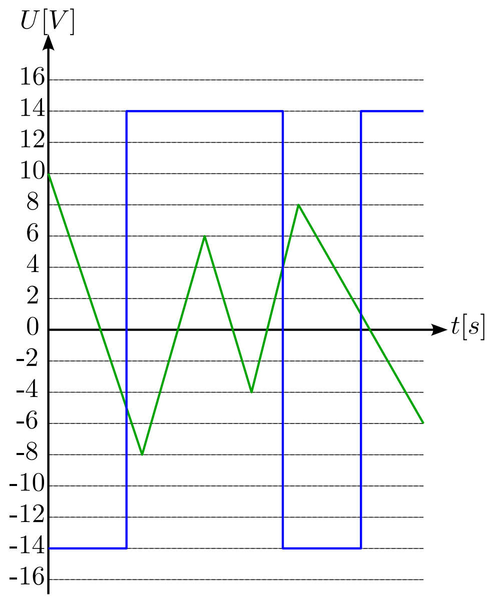
Calculate U_A with the following values:
$R_1 = 10\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$, $R_3 = 30\,\text{k}\Omega$, $R_4 = 10\,\text{k}\Omega$, $R_5 = 20\,\text{k}\Omega$



A.7 Dimensioning of a comparator circuit with diode

Dimension the circuitry of the comparator so that the transfer behaviour outlined below is achieved. The forward voltage of the diode is assumed to be $U_D = 0{,}7\,\text{V}$. The maximum power dissipation of the resistor is $1/4\,\text{W}$.





A.8 Description of the ideal characteristics of an operational amplifier

Describe the following characteristics of an ideal operational amplifier:

a) Input resistance

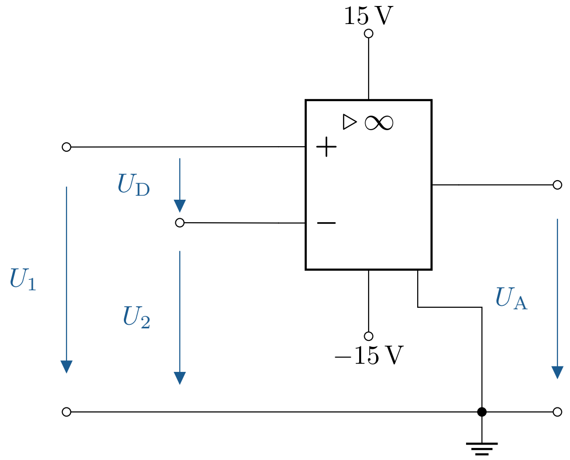
b) Output resistance

c) Differential amplification

d) Input offset voltage

A.9 Determining the output voltage of a comparator

The depicted comparator is given. The voltage $U_1 = 8.25\,\mathrm{V}$ is constant. What is the output voltage for $U_2 = 0\,\mathrm{V}; 4\,\mathrm{V}; 8\,\mathrm{V}; 8.2\,\mathrm{V}; 8.249\,\mathrm{V}; 8.251\,\mathrm{V}; 8.3\,\mathrm{V}; 10\,\mathrm{V}$?



A.10 Calculate comparator circuit for battery monitoring

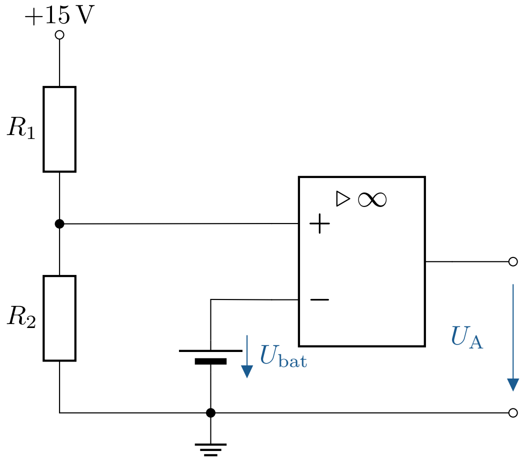
A comparator is to be used to monitor the charge status of a battery. The comparator should switch over when the battery voltage U_{bat} falls below 6 V. The resistor R_1 is 10 kΩ. The operating voltage of the comparator is ±15 V.

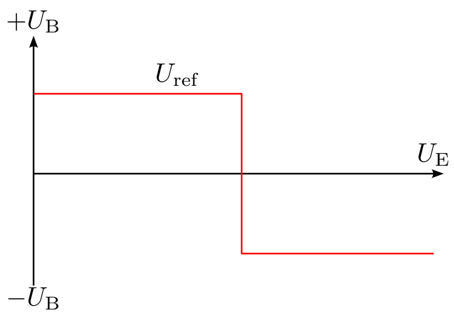
a) What value must the resistor R_2 have?

b) To indicate the charge level, an LED is to be used. Extend the circuit to include a charge indicator using an LED.

c) A maximum current of 20 mA may flow through the LED with a forward voltage of 2.2 V. What value must the series resistor have?

d) The voltage at the voltage divider depends on the supply voltage. What other suitable components exist for generating the reference voltage? Name and draw an example.





A.11 Circuit design of an operational amplifier with two input signals

The following functional relationship exists between the output voltage U_{a} and the input voltages U_1 and U_2: $U_{\mathrm{a}} = -5\ U_1 - 0.8\ U_2$.

Design a circuit using operational amplifier(s) and resistors to realize this functional relationship. Assume the operational amplifiers are ideal. Specify the values of the resistors used!

A.12 Design of an amplifier circuit with different input gains

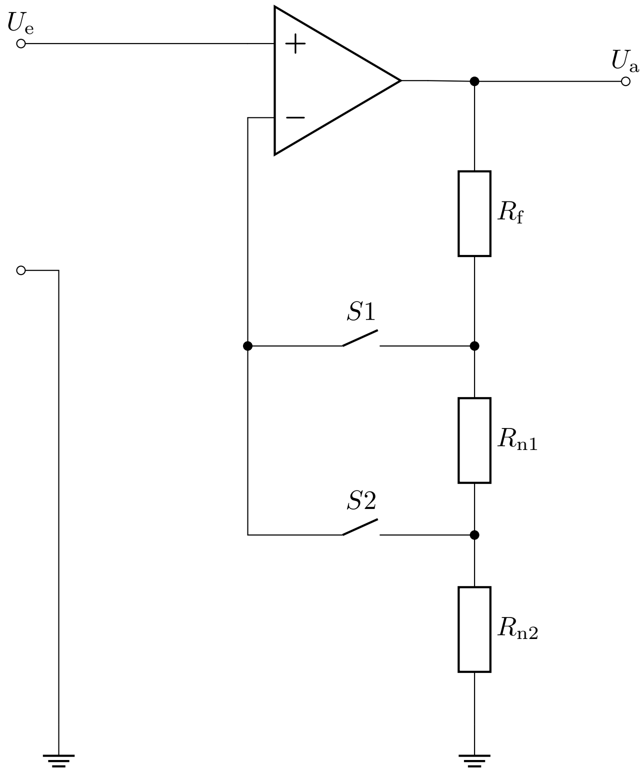
The following functional relationship exists between the output voltage U_{a} and the input voltages U_1 and U_2: $U_{\mathrm{a}} = 2\ U_1 - 0.5\ U_2$.

Design a circuit using operational amplifier(s) and resistors to realize this functional relationship. Assume the operational amplifiers are ideal. Specify the values of the resistors used! Typical resistor values for op-amp circuits are in the range of 1 to 100 kΩ.

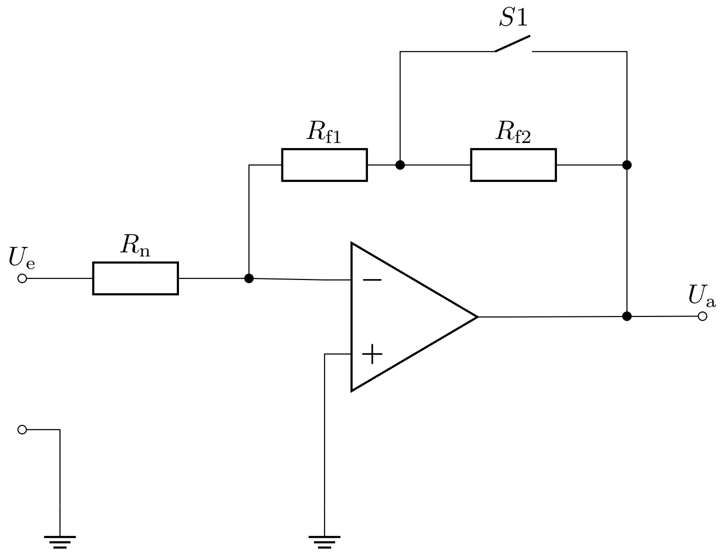
A.13 Gain adjustment with switchable resistors

To adjust the amplification of an operational amplifier, switches can be used to turn various resistors on and off. Calculate the operational amplifier circuits below depending on the switch positions S!

a) For the circuit in Figure 1 (4 switch positions)



b) For the circuit in Figure 2 (2 switch positions)



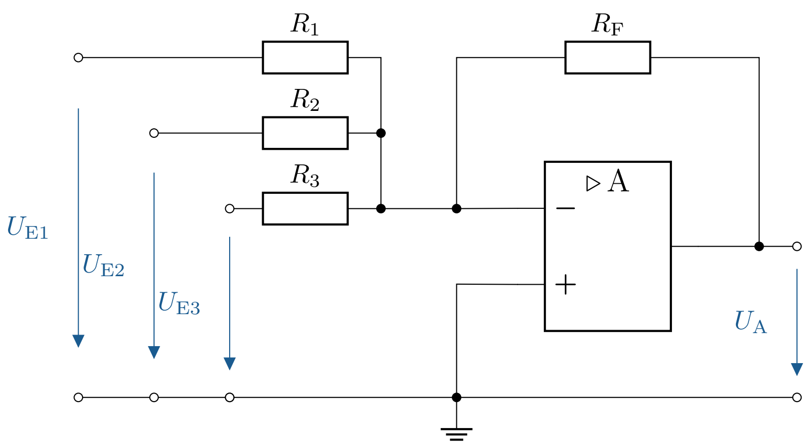
A.14 Calculation of the output voltage with multiple input voltages

The circuit shown with an ideal operational amplifier is given. The operating voltage is $\pm 15\,\mathrm{V}$.

a) All resistors are equal, with $R_1 = R_2 = R_3 = R_F = 1\,\mathrm{k\Omega}$. The input voltages are $U_{E1} = 1\,\mathrm{V}$, $U_{E2} = 2\,\mathrm{V}$, $U_{E3} = 3\,\mathrm{V}$. What is the output voltage U_A?

b) The input voltages remain the same, but the resistors are now $R_F = 10\,\mathrm{k\Omega}$, $R_1 = 2\,\mathrm{k\Omega}$, $R_2 = 5\,\mathrm{k\Omega}$, $R_3 = 10\,\mathrm{k\Omega}$. What is the output voltage U_A?

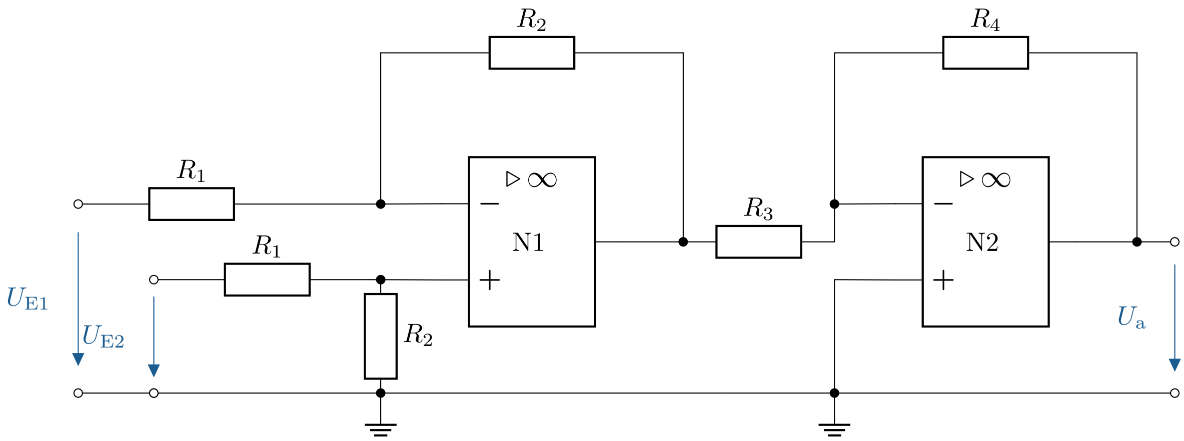
c) The input voltages remain the same, but the resistors are now $R_F = 10\,\mathrm{k\Omega}$, $R_1 = 10\,\mathrm{k\Omega}$, $R_2 = 5\,\mathrm{k\Omega}$, $R_3 = 2\,\mathrm{k\Omega}$. What is the output voltage U_A?



A.15 Determine the output voltage of a two-stage operational amplifier circuit

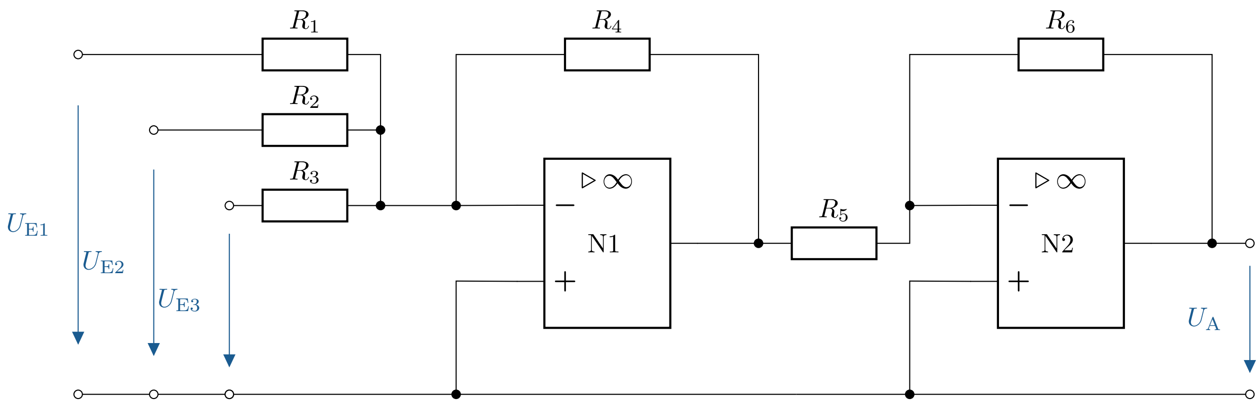
Given is the following circuit. Calculate the output voltage U_a.
The resistance values are:
$R_1 = 10\,\mathrm{k\Omega}$,
$R_2 = 30\,\mathrm{k\Omega}$,

$R_3 = 30\,\mathrm{k}\Omega$,
$R_4 = 10\,\mathrm{k}\Omega$



A.16 Calculation of an OP circuit with three inputs

The circuit shown below is given. Determine the output voltage U_{A}.
The resistance values are:
$R_1 = 10\,\mathrm{k}\Omega$,
$R_2 = 20\,\mathrm{k}\Omega$,
$R_3 = 30\,\mathrm{k}\Omega$,
$R_4 = 10\,\mathrm{k}\Omega$,
$R_5 = 20\,\mathrm{k}\Omega$,
$R_6 = 40\,\mathrm{k}\Omega$



A.17 Investigation of the amplification of an OP with variable resistors

Investigate the amplification of a non-inverting operational amplifier with variable resistors and calculate the output voltage U_{A}. Given:

- Supply voltage: $\pm 15\,\mathrm{V}$
- Input voltage: $U_{\mathrm{E}} = 1\,\mathrm{V}$
- Feedback resistance: $R_{\mathrm{R}} = 100\,\Omega$

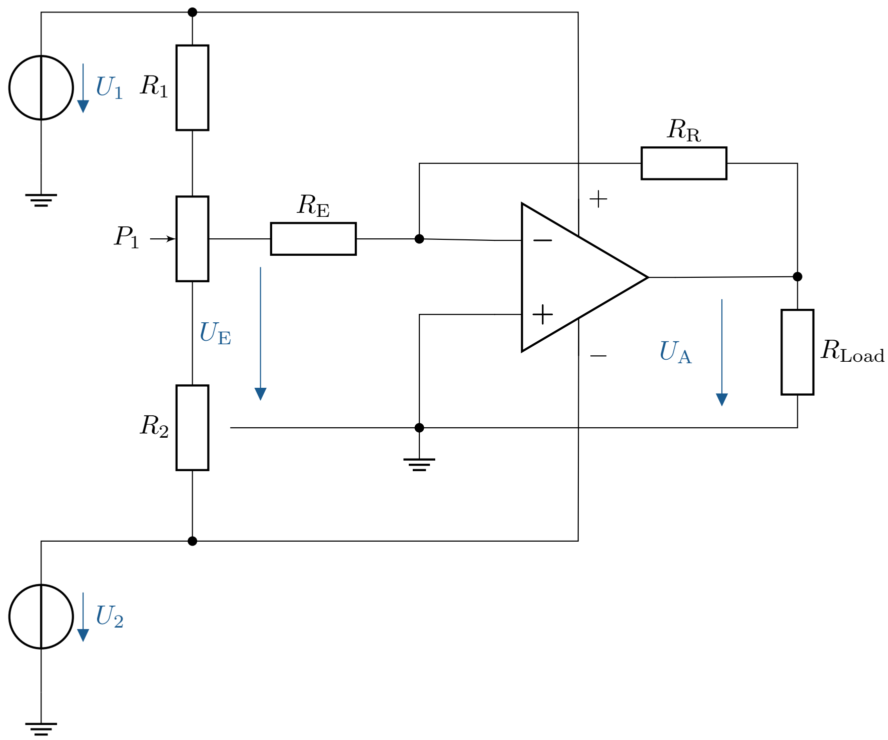
- Input resistance: $R_\mathrm{E} = 680\,\Omega$
- Further possible resistance values: $R_\mathrm{R} = 220\,\Omega$, $R_\mathrm{E} = 680\,\Omega$
- Load resistance: $R_\mathrm{Last} = 10\,\mathrm{k}\Omega$

a) Calculate the gain V and the output voltage U_A for the given resistor values.

b) How does the gain change if R_R is increased from $100\,\Omega$ to $220\,\Omega$?

c) What effects does a change in R_E have on the gain and the output voltage?

d) What limitations for U_A result from the supply voltage?



A.18 Analysis of a non-inverting amplifier with load

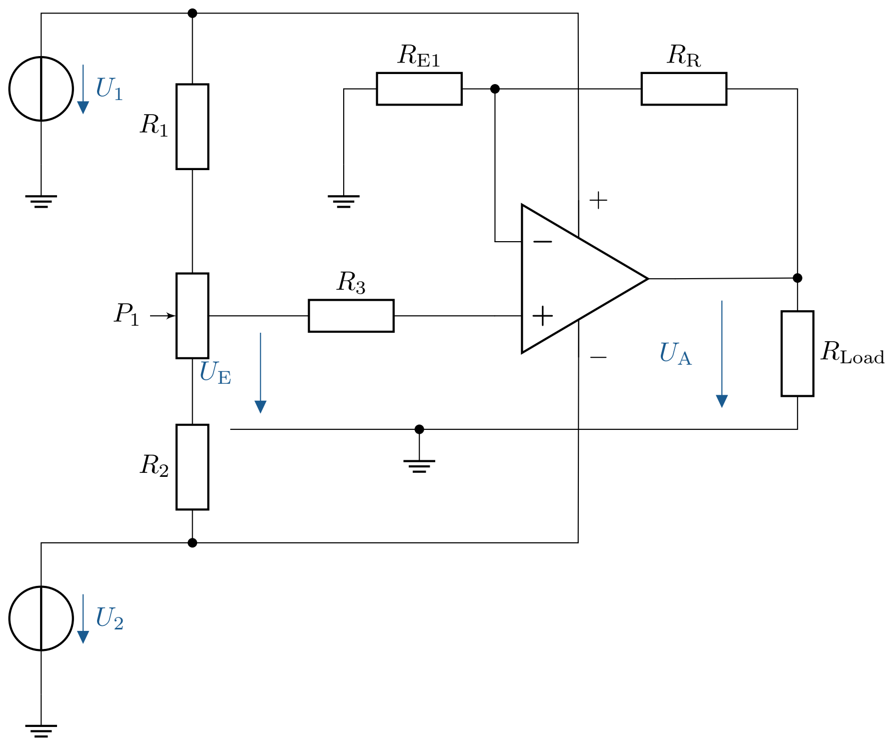
Analyse the gain of a non-inverting operational amplifier and calculate the output voltage U_A. Given:

- Supply voltage: $\pm 15\,\mathrm{V}$
- Input voltage: $U_\mathrm{E} = 1\,\mathrm{V}$
- Feedback resistance: $R_\mathrm{R} = 20\,\mathrm{k}\Omega$
- Input resistance: $R_\mathrm{E} = 10\,\mathrm{k}\Omega$
- Load resistance: $R_\mathrm{Last} = 10\,\mathrm{k}\Omega$

a) Calculate the output voltage U_A using the given values.

b) How does U_A change if R_R is increased to $30\,\mathrm{k}\Omega$?

c) What influence does the load resistor R_Last have on the output signal?



A.19 Analysis of a loaded voltage divider

Draw a circuit diagram consisting of a loaded voltage divider. The component values are given as follows:

- Resistance: $R_1 = R_2 = 10\,k\Omega$
- Load resistance: $R_L = 2{,}2\,\mathrm{k}\Omega$
- Supply voltage: $U_{CC} = +15\,\mathrm{V}$

a) Calculate the output voltage U_C without load resistance.

b) Calculate the output voltage U_C with the connected load resistance. $\mathrm{R_L}$.

c) Explain why the voltage U_C drops across the load resistor.

A.20 Voltage divider with operational amplifier as voltage follower

Draw a new circuit diagram consisting of a loaded voltage divider supplemented by an operational amplifier as an impedance converter. The component values are given as follows:

- Resistances: $R_1 = R_2 = 10\,\mathrm{k}\Omega$
- Load resistance: $R_L = 2{,}2\,\mathrm{k}\Omega$
- Supply voltage: $\mathrm{U_{CC}} = +15\,\mathrm{V}$

The operational amplifier is connected as an impedance converter (voltage follower). Its output feeds the load resistor. R_L.

a) Calculate the output voltage U_C with and without the load resistor R_L.

b) Compare the results with those from Exercise 13.

c) Explain why the voltage U_C does not change due to the load resistor in this circuit.

A.21 Investigation of a step-up converter for voltage boosting

An upward converter can be used to convert small DC voltages (e.g. from a battery) into a higher DC voltage. Examine how an upward converter works by analysing the circuit.

Gegeben:

- Input voltage: $U_E = 1\,\mathrm{V}$
- inductance: $L = 900\,\mu\mathrm{H}$
- Diode: Ideal diode (no voltage drops taken into account)
- capacitor: $C = 470\,\mu\mathrm{F}$
- Switch: Opens and closes periodically

a) Explain how the step-up converter works in two phases:

- *Phase 1: Switch closed*
- *Phase 2: Switch open*

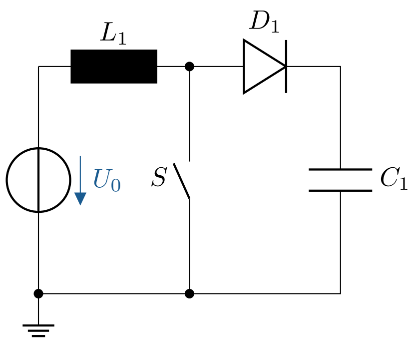
b) Derive the formula for the output voltage U_A as a function of the duty cycle D:

$$U_A = \frac{U_E}{1 - D}$$

c) Calculate U_A for different duty cycles D:

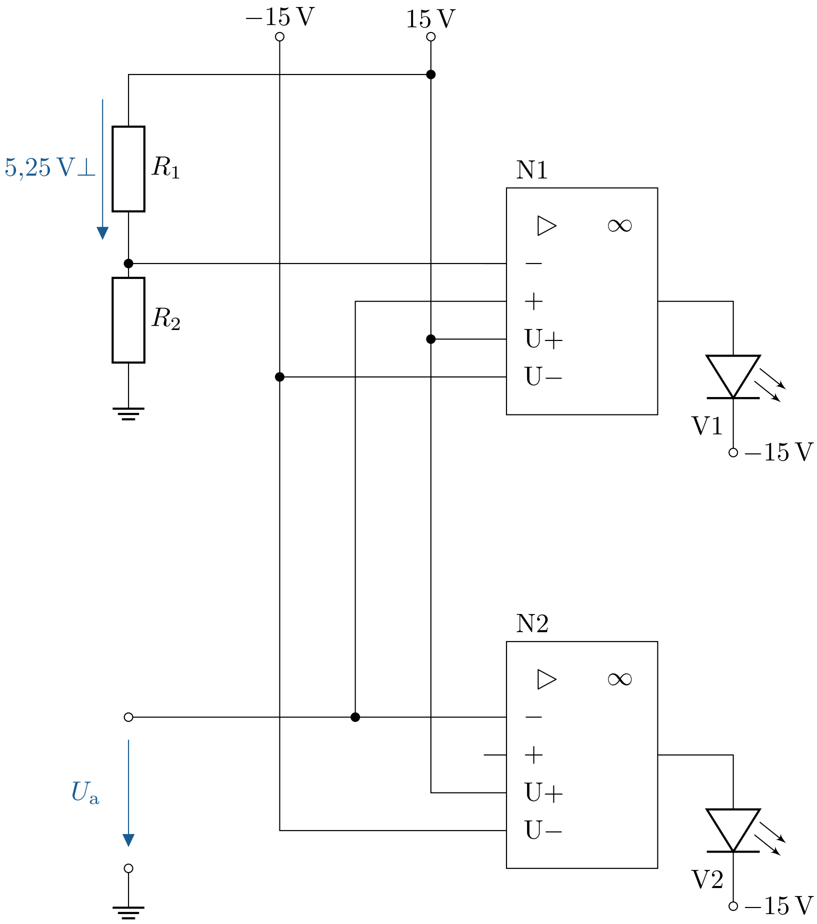
- $D = 0{,}3$
- $D = 0{,}5$
- $D = 0{,}7$

d) What happens when D approaches 1? What practical limitations are there?



B Solutions to the exercises

B.1 Monitoring a voltage with LEDs



$$\frac{U_1}{U_0} = \frac{R_1}{R_1 + R_2}$$

$$\frac{U_1 \cdot (R_1 + R_2)}{U_0} = R_1$$

$$U_1 \cdot (R_1 + R_2) = R_1 \cdot U_0$$

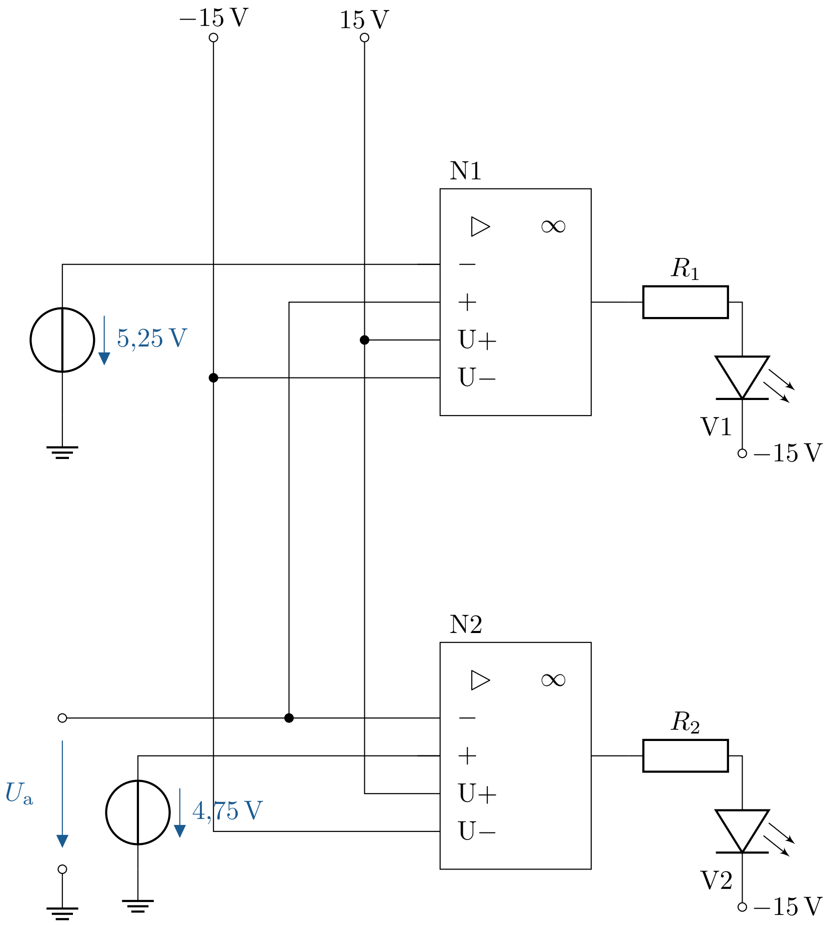
$$U_1 \cdot R_1 + U_1 \cdot R_2 = R_1 \cdot U_0$$

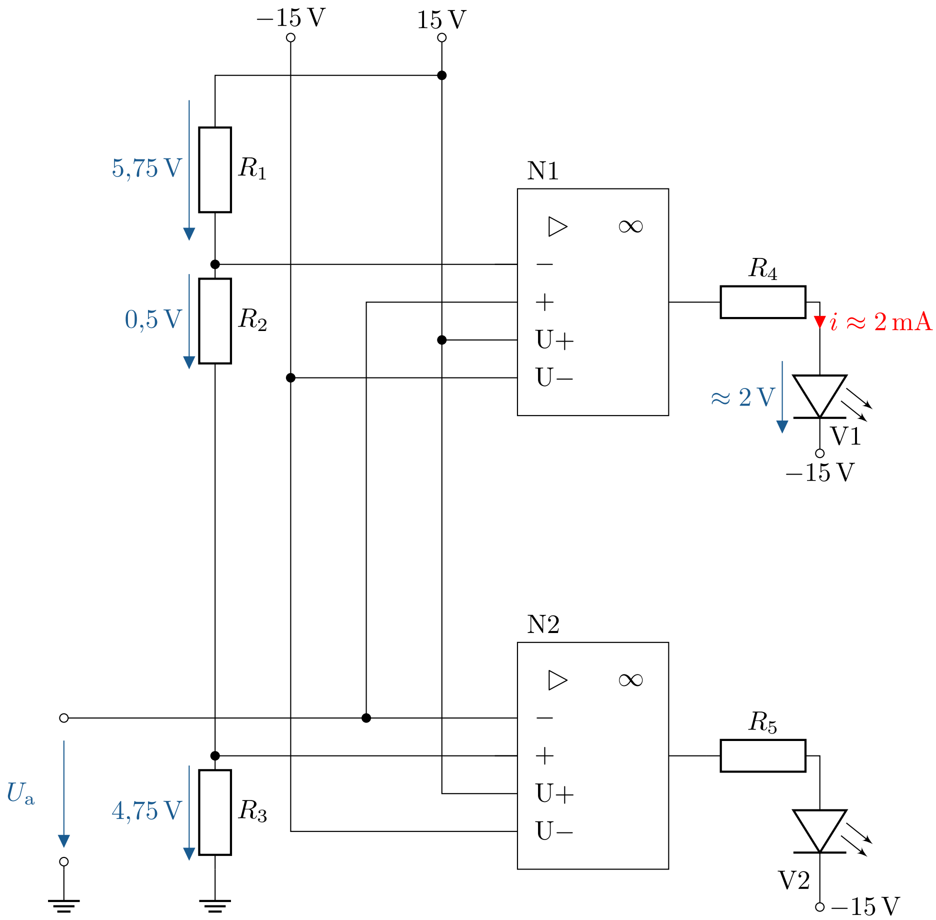
$$U_1 \cdot R_1 = R_1 \cdot U_0 - U_1 \cdot R_2$$

$$U_1 \cdot R_2 = R_1 \cdot (U_0 - U_1)$$

$$\frac{U_1 \cdot R_2}{U_0 - U_1} = R_1$$

$$R_1 = 268.96\,\Omega$$


−15 V
15 V
N1
R1
5,25 V
U+
U−
V1
−15 V
N2
R2
Ua
4,75 V
U+
U−
V2
−15 V



$$
\begin{aligned}
R_1 &= 3.75\,\mathrm{k\Omega}\\
R_2 &= 500\,\Omega\\
R_3 &= 4.75\,\mathrm{k\Omega}\\
R_4 &= \frac{28\,\mathrm{V}}{20\,\mathrm{mA}} \approx 1{,}4\,\mathrm{k\Omega}
\end{aligned}
$$

See section: 3.2 and Table 3.1

B.2 Resistance calculation for non-inverting amplifiers

According to the formula for non-inverting amplifiers:

$$V = 1 + \frac{R_1}{R_2} \tag{B.1}$$

Rearrange the formula according to R_1:

$$R_1 = R_2 \cdot (V - 1) \tag{B.2}$$

- $V = 10$

$$R_1 = 1\,\mathrm{k\Omega} \cdot (10 - 1) = 9\,\mathrm{k\Omega}$$

- $V = 50$

$$R_1 = 1\,\mathrm{k\Omega} \cdot (50 - 1) = 49\,\mathrm{k\Omega}$$

- $V = 100$

$$R_1 = 1\,\mathrm{k\Omega} \cdot (100 - 1) = 99\,\mathrm{k\Omega}$$

See section: 3.2 and Table 3.1

B.3 Resistance calculation for inverting amplifier

According to the formula for inverting amplifiers:

$$V = -\frac{R_2}{R_1} \tag{B.3}$$

Rearrange formula according to R_1:

$$R_1 = -\frac{R_2}{V} \tag{B.4}$$

- $V = -10$

$$R_1 = -\frac{1\,\mathrm{k\Omega}}{-10} = 100\,\Omega$$

- $V = -50$

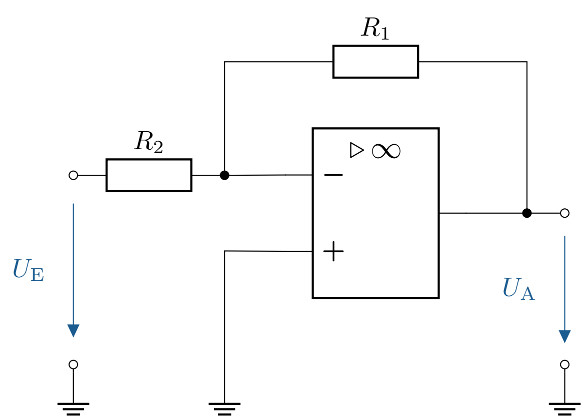
$$R_1 = -\frac{1\,\mathrm{k\Omega}}{-50} = 20\,\Omega$$

- $V = -100$

$$R_1 = -\frac{1\,\mathrm{k\Omega}}{-100} = 10\,\Omega$$

See section: 3.2 and Table 3.1

B.4 Calculation of the output voltage of an operational amplifier circuit with two inputs



Calculate reinforcement:

$$V = -\frac{100\,\mathrm{k\Omega}}{10\,\mathrm{k\Omega}} = -10$$

Formula for amplification:

$$V = \frac{U_\mathrm{A}}{U_\mathrm{E}} \tag{B.5}$$

Rearrange formula according to U_A:

$$U_\mathrm{A} = V \cdot U_\mathrm{E}$$

a)

$$U_\mathrm{A} = -10 \cdot 2\,\mathrm{mV} = -20\,\mathrm{mV}$$

b)

$$U_\mathrm{A} = -10 \cdot 20\,\mathrm{mV} = -200\,\mathrm{mV}$$

See section: 3.2 and Table 3.1

B.5 Calculate the output voltage of an operational amplifier with multiple inputs

Formula:

$$U_\mathrm{A} = -(U_\mathrm{E1} \cdot \frac{R_\mathrm{F}}{R_1} + U_\mathrm{E2} \cdot \frac{R_\mathrm{F}}{R_2} + U_\mathrm{E3} \cdot \frac{R_\mathrm{F}}{R_3}) \tag{B.6}$$

a)

$$U_\mathrm{A} = -(1\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{10\,\mathrm{k\Omega}} + 2\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{10\,\mathrm{k\Omega}} + 3\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{10\,\mathrm{k\Omega}}) = -6\,\mathrm{V}$$

b)

$$U_\mathrm{A} = -(1\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{2\,\mathrm{k\Omega}} + 2\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{5\,\mathrm{k\Omega}} + 3\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{10\,\mathrm{k\Omega}}) = -12\,\mathrm{V}$$

c)

$$U_\mathrm{A} = -(1\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{10\,\mathrm{k\Omega}} + 2\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{5\,\mathrm{k\Omega}} + 3\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{2\,\mathrm{k\Omega}}) = -20\,\mathrm{V}$$

Note: $-20\,\mathrm{V}$ is not possible because the supply voltage is $\pm 15\,\mathrm{V}$. This means the lowest possible value is $-15\,V$.

See section: 3.2 and Table 3.1

B.6 Calculation of the output voltage of a two-stage operational amplifier circuit

Formula:

$$U_\mathrm{N1} = -\left(\frac{R_\mathrm{N}}{R_1} U_\mathrm{E1} + \frac{R_\mathrm{N}}{R_2} U_\mathrm{E2} + \frac{R_\mathrm{N}}{R_3} U_\mathrm{E3}\right) \tag{B.7}$$

$$U_\mathrm{A} = -\frac{R_6}{R_5} U_\mathrm{N1} \tag{B.8}$$

Calculation:

Calculation of U_N1

$$\begin{aligned} U_\mathrm{N1} &= -\left(\frac{10\,\mathrm{k\Omega}}{10\,\mathrm{k\Omega}} U_\mathrm{E1} + \frac{10\,\mathrm{k\Omega}}{20\,\mathrm{k\Omega}} U_\mathrm{E2} + \frac{10\,\mathrm{k\Omega}}{30\,\mathrm{k\Omega}} U_\mathrm{E3}\right) \\ &= -\left(U_\mathrm{E1} + 0{,}5 U_\mathrm{E2} + \frac{1}{3} U_\mathrm{E3}\right) \end{aligned}$$

Calculation of U_A

$$\begin{aligned} U_\mathrm{A} &= -2 \cdot U_\mathrm{N1} \\ &= -2\left(-\left(U_\mathrm{E1} + 0{,}5 U_\mathrm{E2} + \frac{1}{3} U_\mathrm{E3}\right)\right) \\ &= 2\left(U_\mathrm{E1} + 0{,}5 U_\mathrm{E2} + \frac{1}{3} U_\mathrm{E3}\right) \end{aligned}$$

The output voltage U_A is therefore:

$$U_\mathrm{A} = 2U_\mathrm{E1} + U_\mathrm{E2} + \frac{2}{3} U_\mathrm{E3}$$

See section: 3.2 and Table 3.1

B.7 Dimensioning of a comparator circuit with diode

Formula:

$$P_\mathrm{R} = \frac{U_\mathrm{R}^2}{R} \tag{B.9}$$

Calculation:

Voltage across the resistor R:

$$\begin{aligned} U_\mathrm{R} &= U_\mathrm{a} - U_\mathrm{D} \\ &= 14\,\mathrm{V} - 0{,}7\,\mathrm{V} \\ &= 13{,}3\,\mathrm{V} \end{aligned}$$

Calculation of resistance R:

$$\begin{aligned} P_\mathrm{R} &= \frac{U_\mathrm{R}^2}{R} \leq \frac{1}{4}\,\mathrm{W} \\ R &= \frac{U_R^2}{P_\mathrm{max}} \\ &= \frac{(13{,}3\,\mathrm{V})^2}{0{,}25\,\mathrm{W}} \\ &= \frac{176{,}89\,\mathrm{V}^2}{0{,}25\,\mathrm{W}} \\ &= 707{,}56\,\Omega \end{aligned}$$

Formed resistance value:

$$R = 710\,\Omega$$

Checking the power loss:

$$P_{\mathrm{R}} = \frac{(13{,}3\,\mathrm{V})^2}{710\,\Omega} \approx 0{,}25\,\mathrm{W}$$

The resistance R should have the value:

$$R = 710\,\Omega$$

In order not to exceed the maximum power dissipation and to ensure the desired transmission behaviour. See section: 3.2 and Table 3.1

B.8 Description of the ideal characteristics of an operational amplifier

a) Input resistance: infinite, to minimize loading on the voltage source

b) Output resistance: $0\,\Omega$, so that the op-amp can deliver as much power as possible

c) Differential gain: infinite, to amplify small signals

d) Input offset voltage: $0\,\mathrm{V}$, so that the op-amp produces no output voltage when there is no input signal

See section: 2.1

B.9 Determining the output voltage of a comparator

Exploitation of the mesh rule states:

$$\begin{aligned} -U_1 + U_2 + U_D &= 0 \\ U_{\mathrm{D}} &= U_1 - U_2 \end{aligned}$$

- $U_2 = 0\,V$

$$\begin{aligned} U_{\mathrm{D}} &= 8{,}25\,\mathrm{V} - 0\,\mathrm{V} = 8{,}25\,\mathrm{V} \\ U_{\mathrm{A}} &= 13{,}5\,\mathrm{V} \end{aligned}$$

- $U_2 = 4\,V$

$$\begin{aligned} U_{\mathrm{D}} &= 8{,}25\,\mathrm{V} - 4\,\mathrm{V} = 4{,}25\,\mathrm{V} \\ U_{\mathrm{A}} &= 13{,}5\,\mathrm{V} \end{aligned}$$

- $U_2 = 8\,V$

$$\begin{aligned} U_{\mathrm{D}} &= 8{,}25\,\mathrm{V} - 8\,\mathrm{V} = 0{,}25\,\mathrm{V} \\ U_{\mathrm{A}} &= 13{,}5\,\mathrm{V} \end{aligned}$$

- $U_2 = 8{,}2\,V$

$$\begin{aligned} U_{\mathrm{D}} &= 8{,}25\,\mathrm{V} - 8{,}2\,\mathrm{V} = 0{,}05\,\mathrm{V} \\ U_{\mathrm{A}} &= 8\,\mathrm{V} \end{aligned}$$

- $U_2 = 8{,}249\,V$

$$\begin{aligned} U_{\mathrm{D}} &= 8{,}25\,\mathrm{V} - 8{,}249\,\mathrm{V} = 1\,\mathrm{mV} \\ U_{\mathrm{A}} &= 4\,\mathrm{V} \end{aligned}$$

- $U_2 = 8{,}251\,V$

$$U_D = 8{,}25\,\mathrm{V} - 8{,}251\,\mathrm{V} = -1\,\mathrm{mV}$$
$$U_A = -4\,\mathrm{V}$$

- $U_2 = 8{,}3\,V$

$$U_D = 8{,}25\,\mathrm{V} - 8{,}3\,\mathrm{V} = -0{,}05\,\mathrm{V}$$
$$U_A = -8\,\mathrm{V}$$

- $U_2 = 10\,V$

$$U_D = 8{,}25\,\mathrm{V} - 10\,\mathrm{V} = -1{,}75\,\mathrm{V}$$
$$U_A = -13{,}5\,\mathrm{V}$$

See section: 3.2 and Table 3.1

B.10 Calculate comparator circuit for battery monitoring

a) Calculation of resistance R_2

The comparator should switch when the battery voltage U_{bat} falls below 6 V. The ratio of resistors R_1 and R_2 in the voltage divider must be selected so that the reference voltage $U_{ref} = 6\,\mathrm{V}$ is applied to the non-inverting input:

$$U_{ref} = \frac{R_2}{R_1 + R_2} \cdot U_B$$

with $U_B = 15\,\mathrm{V}$ and $R_1 = 10\,\mathrm{k\Omega}$. Insert:

$$6\,\mathrm{V} = \frac{R_2}{10\,\mathrm{k\Omega} + R_2} \cdot 15\,\mathrm{V}$$

Convert to R_2:

$$R_2 = \frac{6\,\mathrm{V} \cdot 10\,\mathrm{k\Omega}}{15\,\mathrm{V} - 6\,\mathrm{V}} = 6{,}66\,\mathrm{k\Omega}$$

b) Adding an LED display to the circuit:

The LED should indicate when the battery voltage falls below 6 V. To do this, the LED is connected to the output of the comparator with a series resistor R_V.

- Switching behaviour of the circuit:
  - **Battery empty ($U_{bat} = 5\,\mathrm{V}$):**

$$U_+ = 6\,\mathrm{V} - U_{bat}$$
$$U_+ = 6\,\mathrm{V} - 5\,\mathrm{V} = 1\,\mathrm{V}$$

Since $U_+ < U_{ref}$, the comparator switches:

$$U_A = -15\,\mathrm{V}$$

The LED is lit.

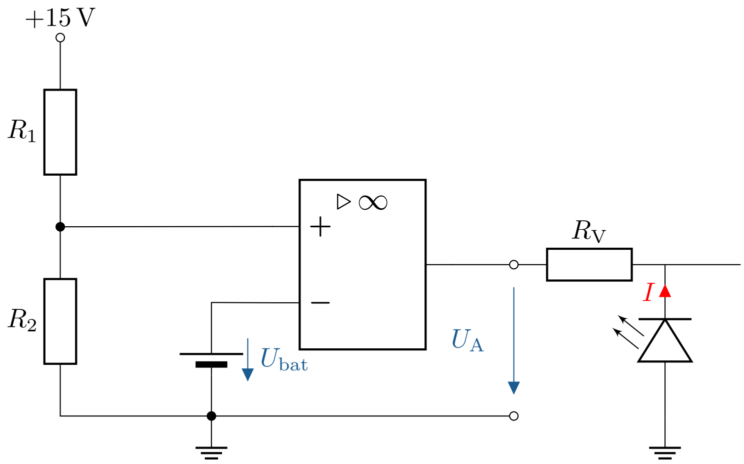
  - **Full battery ($U_{bat} = 7\,\mathrm{V}$):**

$$U_+ = 6\,\mathrm{V} - U_{bat}$$
$$U_+ = 6\,\mathrm{V} - 7\,\mathrm{V} = -1\,\mathrm{V}$$

Since $U_+ > U_{ref}$, the comparator switches:

$$U_A = 15\,\mathrm{V}$$

The LED remains off.



c) Dimensioning of the series resistor R_{V}

Given:

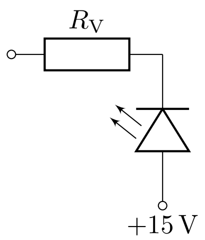
- LED forward voltage: $U_{\text{F}} = 2{,}2\,\text{V}$
- Current through the LED: $I_{\text{F}} = 20\,\text{mA}$
- Output voltage of the comparator: $U_{\text{A}} = 15\,\text{V}$

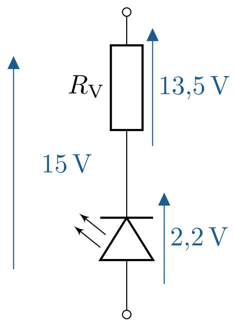
The series resistorR_{V} is calculated from:

$$R_{\text{V}} = \frac{U_{\text{A}} - U_{\text{F}}}{I_{\text{F}}}$$

Applying the values:

$$R_{\text{V}} = \frac{15\,\text{V} - 2{,}2\,\text{V}}{20\,\text{mA}} = \frac{13{,}8\,\text{V}}{0{,}02\,\text{A}} = 690\,\Omega$$

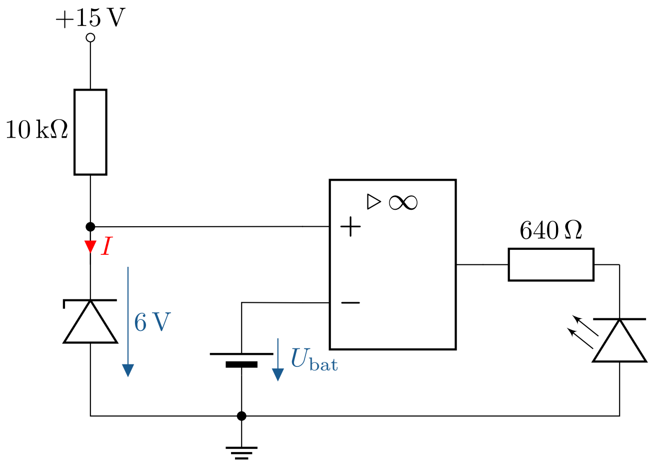




d) Alternative voltage reference:

Since the voltage of the voltage divider depends on the operating voltage, a Zener diode could be used as a reference element. A Zener diode provides a constant reference voltage regardless of fluctuations in the supply voltage.

- Example circuit:
  - The Zener diode is connected to the operating voltage +15 V with a series resistor.
  - The reference voltage is tapped directly from the cathode of the Zener diode.



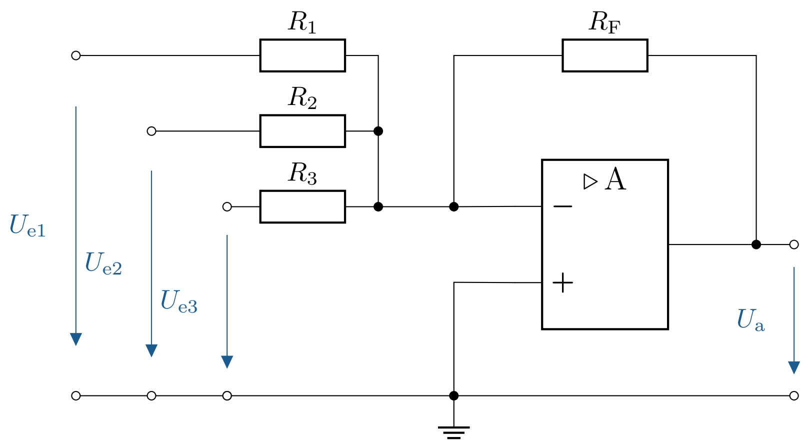
See section: 3.2 and table 3.1

B.11 Circuit design of an operational amplifier with two input signals

The given functional relationship between the output voltage U_{a} and the input voltages U_1 and U_2 is: $U_{\text{a}} = -5U_1 - 0{,}8U_2$

1. Circuit description

To implement this relationship, we use a summing circuit with an inverting operational amplifier. The output voltage of the inverting summing amplifier is: $U_{\text{a}} = -\left(\frac{R_{\text{F}}}{R_1}U_1 + \frac{R_{\text{F}}}{R_2}U_2\right)$ Comparison with the target: $-\left(\frac{R_{\text{F}}}{R_1}\right) = -5 \quad \Rightarrow \quad \frac{R_{\text{F}}}{R_1} = 5$ $-\left(\frac{R_{\text{F}}}{R_2}\right) = -0{,}8 \quad \Rightarrow \quad \frac{R_{\text{F}}}{R_2} = 0{,}8$



2. Selection of resistance values

We set $R_{\text{F}} = 10\,\text{k}\Omega$ (typical value in the range 1–100 kΩ):

- **Calculation of R_1:** $\frac{R_{\text{F}}}{R_1} = 5 \quad \Rightarrow \quad R_1 = \frac{R_{\text{F}}}{5} = \frac{10\,\text{k}\Omega}{5} = 2\,\text{k}\Omega$
- **Calculation of R_2:** $\frac{R_{\text{F}}}{R_2} = 0{,}8 \quad \Rightarrow \quad R_2 = \frac{R_{\text{F}}}{0{,}8} = \frac{10\,\text{k}\Omega}{0{,}8} = 12{,}5\,\text{k}\Omega$

3. Implementation of the circuit

The resistors used:

$$R_F = 10\,\mathrm{k\Omega}$$
$$R_1 = 2\,\mathrm{k\Omega}$$
$$R_2 = 12{,}5\,\mathrm{k\Omega}$$

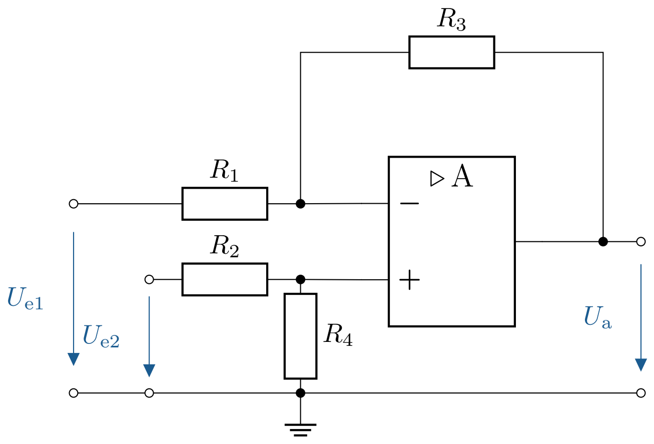
4. Circuit diagram of the implemented circuit

The circuit consists of an operational amplifier whose inverting input is connected to resistors R_1 and R_2, which feed in input voltages U_1 and U_2. The non-inverting input is connected to ground.

Summary:

The desired functional equation $U_a = -5U_1 - 0.8U_2$ is realised using the circuit described and the calculated resistance values. See section: 3.2 and Table 3.1

B.12 Design of an amplifier circuit with different input gains



The functional relationship between the output voltage U_a and the input voltages U_{e1} and U_{e2} is as follows:

$$U_a = 2U_{e1} - 0{,}5U_{e2}$$

1. General formula for the output voltage of a subtractor:

$$U_a = U_{e2} \cdot \frac{R_1 + R_3}{R_2 + R_4} \cdot \frac{R_4}{R_1} - U_{e1} \cdot \frac{R_3}{R_1}$$

Comparing with the specification $U_a = 2U_{e1} - 0.5U_{e2}$ gives the following conditions:
- Amplification factor for U_{e1}: $\frac{R_3}{R_1} = 0.5$
- Amplification factor for U_{e2}: $\frac{R_1+R_3}{R_2+R_4} \cdot \frac{R_4}{R_1} = 2$

2. Procedure:
- Set $R_3 = 1\,\mathrm{k\Omega}$ (typical value).
- Calculate the remaining resistances..

3. Calculation of resistances:

Calculation of R_1 (from $\frac{R_3}{R_1} = 0{,}5$):

$$\frac{R_3}{R_1} = 0{,}5 \quad \Rightarrow \quad R_1 = \frac{R_3}{0{,}5} = \frac{1\,\mathrm{k\Omega}}{0{,}5} = 2\,\mathrm{k\Omega}$$

Calculation ofR_2 **and** R_4 **(from** $\frac{R_1+R_3}{R_2+R_4} \cdot \frac{R_4}{R_1} = 2$**):** Set $R_4 = 1\,\mathrm{k\Omega}$:

$$\frac{R_1 + R_3}{R_2 + R_4} \cdot \frac{R_4}{R_1} = 2 \quad \Rightarrow \quad \frac{3\,\mathrm{k\Omega}}{R_2 + 1\,\mathrm{k\Omega}} \cdot \frac{1\,\mathrm{k\Omega}}{2\,\mathrm{k\Omega}} = 2$$

Multiply both sides:

$$\frac{3}{2} = 2 \cdot (R_2 + 1\,\mathrm{k\Omega})$$

Convert to R_2:

$$R_2^2 + R_2 - \frac{3}{2} = 0$$

4. Solution using the PQ formula:

The quadratic equation has the form:

$$R_2^2 + p \cdot R_2 + q = 0 \quad \text{mit} \quad p = 1,\, q = -\frac{3}{2}$$

PQ formula:

$$R_2 = -\frac{p}{2} \pm \sqrt{\left(\frac{p}{2}\right)^2 - q}$$

Applying the values:

$$R_2 = -\frac{1}{2} \pm \sqrt{\left(\frac{1}{2}\right)^2 - \left(-\frac{3}{2}\right)} = -\frac{1}{2} \pm \sqrt{\frac{1}{4} + \frac{3}{2}}$$

$$R_2 = -\frac{1}{2} \pm \sqrt{\frac{7}{4}} = -\frac{1}{2} \pm \frac{\sqrt{7}}{2}$$

Since R_2 must be positive:

$$R_2 = -\frac{1}{2} + \frac{\sqrt{7}}{2} \approx 1{,}82\,\mathrm{k\Omega}$$

5. Result of the resistance:

$$R_1 = 2\,\mathrm{k\Omega}$$
$$R_2 = 1{,}82\,\mathrm{k\Omega}$$
$$R_3 = 1\,\mathrm{k\Omega}$$
$$R_4 = 1\,\mathrm{k\Omega}$$

See section: 3.2 and Table 3.1

B.13 Gain adjustment with switchable resistors

a) Non-inverting amplifier The gain can be changed using two switches (S1 and S2), resulting in four possible combinations:

Case 1 (S1 closed, S2 open): In this state, resistors R_{n1} and R_{n2} form a series connection. The total gain is therefore:

$$V = 1 + \frac{R_{\mathrm{f}}}{R_{\mathrm{n1}} + R_{\mathrm{n2}}}$$

Case 2 (S1 open, S2 closed): Here, resistor R_{f} and R_{n1} are in series. This series combination, together with resistor R_{n2}, forms the feedback loop. The gain is therefore:

$$V = 1 + \frac{R_{\mathrm{f}} + R_{\mathrm{n1}}}{R_{\mathrm{n2}}}$$

Case 3 (S1 and S2 closed): With both switches closed, resistor R_{n1} is shorted and thus not considered. The gain is:

$$V = 1 + \frac{R_{\mathrm{f}}}{R_{\mathrm{n2}}}$$

Case 4 (S1 and S2 open): If both switches are open, the feedback path is missing completely, so the gain is theoretically infinite:

$$V = \infty$$

b) Inverting amplifier **Case 1 (S1 open):** If R_{f1} and R_{f2} are in series, this results in more feedback and thus higher gain. The gain is calculated as:

$$V = -\frac{R_{\mathrm{f1}} + R_{\mathrm{f2}}}{R_{\mathrm{n}}}$$

Case 2 (S1 closed, S2 open): In this state, resistor R_{f2} is shorted by the closed switch and not considered. The gain is reduced to:

$$V = -\frac{R_{\mathrm{f1}}}{R_{\mathrm{n}}}$$

See section: 3.2 and Table 3.1

B.14 Calculation of the output voltage with multiple input voltages

Given is a summing circuit with an ideal operational amplifier and an operating voltage of $\pm 15\,\mathrm{V}$.

The general formula for the output voltage is:

$$U_{\mathrm{A}} = -\left(U_{\mathrm{E1}} \cdot \frac{R_{\mathrm{F}}}{R_1} + U_{\mathrm{E2}} \cdot \frac{R_{\mathrm{F}}}{R_2} + U_{\mathrm{E3}} \cdot \frac{R_{\mathrm{F}}}{R_3}\right)$$

a) All resistances are equal in size.: $R_1 = R_2 = R_3 = R_F = 10\,\mathrm{k\Omega}$

Input voltages:

$$U_{\mathrm{E1}} = 1\,\mathrm{V}, \quad U_{\mathrm{E2}} = 2\,\mathrm{V}, \quad U_{\mathrm{E3}} = 3\,\mathrm{V}$$

Calculation:

$$U_{\mathrm{A}} = -\left(1\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{10\,\mathrm{k\Omega}} + 2\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{10\,\mathrm{k\Omega}} + 3\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{10\,\mathrm{k\Omega}}\right)$$
$$U_A = -\left(1\,\mathrm{V} + 2\,\mathrm{V} + 3\,\mathrm{V}\right) = -6\,\mathrm{V}$$

b) Modified resistances: $R_{\mathrm{F}} = 10\,\mathrm{k\Omega}$**,** $R_1 = 2\,\mathrm{k\Omega}$**,** $R_2 = 5\,\mathrm{k\Omega}$**,** $R_3 = 10\,\mathrm{k\Omega}$

Input voltages:

$$U_{\mathrm{E1}} = 1\,\mathrm{V}, \quad U_{\mathrm{E2}} = 2\,\mathrm{V}, \quad U_{\mathrm{E3}} = 3\,\mathrm{V}$$

Calculation:

$$U_{\mathrm{A}} = -\left(1\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{2\,\mathrm{k\Omega}} + 2\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{5\,\mathrm{k\Omega}} + 3\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{10\,\mathrm{k\Omega}}\right)$$
$$U_{\mathrm{A}} = -\left(1\,\mathrm{V} \cdot 5 + 2\,\mathrm{V} \cdot 2 + 3\,\mathrm{V} \cdot 1\right)$$
$$U_{\mathrm{A}} = -(5\,\mathrm{V} + 4\,\mathrm{V} + 3\,\mathrm{V}) = -12\,\mathrm{V}$$

c) Modified resistances: $R_{\mathrm{F}} = 10\,\mathrm{k\Omega}$**,** $R_1 = 10\,\mathrm{k\Omega}$**,** $R_2 = 5\,\mathrm{k\Omega}$**,** $R_3 = 2\,\mathrm{k\Omega}$

Input voltages:

$$U_{\mathrm{E1}} = 1\,\mathrm{V}, \quad U_{\mathrm{E2}} = 2\,\mathrm{V}, \quad U_{\mathrm{E3}} = 3\,\mathrm{V}$$

Calculation:

$$U_\mathrm{A} = -\left(1\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{10\,\mathrm{k\Omega}} + 2\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{5\,\mathrm{k\Omega}} + 3\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{2\,\mathrm{k\Omega}}\right)$$
$$U_\mathrm{A} = -\left(1\,\mathrm{V} \cdot 1 + 2\,\mathrm{V} \cdot 2 + 3\,\mathrm{V} \cdot 5\right)$$
$$U_\mathrm{A} = -(1\,\mathrm{V} + 4\,\mathrm{V} + 15\,\mathrm{V}) = -20\,\mathrm{V}$$

Since the operating voltage is $\pm 15\,\mathrm{V}$, the output is limited to $-15\,\mathrm{V}$.

Summary:

- a) $U_\mathrm{A} = -6\,\mathrm{V}$
- b) $U_\mathrm{A} = -12\,\mathrm{V}$
- c) $U_\mathrm{A} = -15\,\mathrm{V}$ (Limitation by operating voltage)

See section: 3.2 and Table 3.1

B.15 Determine the output voltage of a two-stage operational amplifier circuit

The output voltage after the first operational amplifier (U_a1) is calculated as follows:

$$U_\mathrm{a1} = (U_\mathrm{E1} - U_\mathrm{E2}) \cdot \frac{R_2}{R_1}$$
$$U_\mathrm{a1} = \frac{30\,\mathrm{k\Omega}}{10\,\mathrm{k\Omega}} \cdot (U_\mathrm{E1} - U_\mathrm{E2})$$

This results in the following for the total output voltage (U_a):

$$U_\mathrm{a} = -\frac{R_4}{R_3} \cdot U_\mathrm{a1}$$
$$U_\mathrm{a} = -\frac{10\,\mathrm{k\Omega}}{30\,\mathrm{k\Omega}} \cdot \frac{30\,\mathrm{k\Omega}}{10\,\mathrm{k\Omega}} \cdot (U_\mathrm{E1} - U_\mathrm{E2})$$
$$U_\mathrm{a} = U_\mathrm{E2} - U_\mathrm{E1}$$

See section: 3.2 and Table 3.1

B.16 Calculation of an OP circuit with three inputs

Calculate the output voltage after the first operational amplifier using the following formula (U_{a1}):

$$U_\mathrm{A1} = -(U_\mathrm{E1} \cdot \frac{R_4}{R_1} + U_\mathrm{E2} \cdot \frac{R_4}{R_2} + U_\mathrm{E3} \cdot \frac{R_4}{R_3})$$
$$U_\mathrm{A1} = -(U_\mathrm{E1} \cdot \frac{10}{10} + U_\mathrm{E2} \cdot \frac{10}{20} + U_\mathrm{E3} \cdot \frac{10}{30})$$
$$U_\mathrm{A1} = -(U_\mathrm{E1} + \frac{1}{2} \cdot U_\mathrm{E2} + \frac{1}{3} \cdot U_\mathrm{E3})$$

This results in the following for the total output voltage (U_A):

$$U_\mathrm{A} = -\frac{R_6}{R_5} \cdot U_\mathrm{a1}$$
$$U_\mathrm{A} = -\frac{40}{20} \cdot (-(U_\mathrm{E1} + \frac{1}{2} \cdot U_\mathrm{E2} + \frac{1}{3} \cdot U_\mathrm{E3}))$$
$$U_\mathrm{A} = 2 \cdot U_\mathrm{E1} + U_\mathrm{E2} + \frac{2}{3} \cdot U_\mathrm{E3}$$

See section: 3.2 and Table 3.1

B.17 Investigation of the amplification of an OP with variable resistors

a) Calculation of the output voltage U_A for given values
Given is:

$$U_E = 1\,\text{V}, \quad R_R = 100\,\Omega, \quad R_E = 680\,\Omega$$

Insert into the amplification formula:

$$V = 1 + \frac{100\,\Omega}{680\,\Omega} = 1 + 0{,}147 = 1{,}147$$

The output voltage is calculated as follows:

$$U_A = V \cdot U_E = 1{,}147 \cdot 1\,\text{V} = 1{,}147\,\text{V}$$

b) Change from R_R to $220\,\Omega$

When $R_R = 220\,\Omega$ is selected:

$$V = 1 + \frac{220\,\Omega}{680\,\Omega} = 1 + 0{,}323 = 1{,}323$$

$$U_A = 1{,}323 \cdot 1\,\text{V} = 1{,}323\,\text{V}$$

c) Influence of R_E on the gain

An increase in R_E leads to a smaller gain V, while a decrease in R_E increases the gain. In practice, R_E should be chosen so that the output signal does not fluctuate too much.

d) Limitation of U_A by the supply voltage

The output voltage U_A can never exceed the supply voltage $\pm 15\,\text{V}$. If U_A would exceed this range according to calculations, the output voltage is limited to a maximum of 15 V. See section: 3.2 and Table 3.1

B.18 Analysis of a non-inverting amplifier with load

a) Calculation of the output voltage $\mathrm{U_A}$
Given are:

$$U_E = 1\,\text{V}, \quad R_R = 20\,\text{k}\Omega, \quad R_E = 10\,\text{k}\Omega$$

Insert into the amplification formula:

$$V = 1 + \frac{20\,\text{k}\Omega}{10\,\text{k}\Omega} = 1 + 2 = 3$$

The output voltage is calculated as follows:

$$U_A = V \cdot U_E = 3 \cdot 1\,\text{V} = 3\,\text{V}$$

b) Change from $\mathrm{R_R}$

If R_R is increased to $30\,\text{k}\Omega$:

$$V = 1 + \frac{30\,\text{k}\Omega}{10\,\text{k}\Omega} = 1 + 3 = 4$$

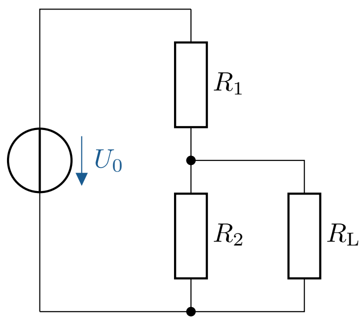
$$U_A = 4 \cdot 1\,\text{V} = 4\,\text{V}$$

c) Influence of load resistance

The load resistance R_{Load} has only a minimal effect on the circuit as long as the operational amplifier is operating in the linear range. However, if the load resistance is too small, the operational amplifier may reach its output current limit, resulting in voltage limitation.

See section: 3.2 and Table 3.1

B.19 Analysis of a loaded voltage divider



a) Calculation of the output voltage U_C without load resistance:

$$U_C = \frac{R_2}{R_1 + R_2} \cdot U_{CC}$$
$$U_C = \frac{10\,k\Omega}{10\,k\Omega + 10\,\text{k}\Omega} \cdot 15\,\text{V}$$
$$U_C = \frac{1}{2} \cdot 15\,\text{V}$$
$$U_C = 7{,}5\,\text{V}$$

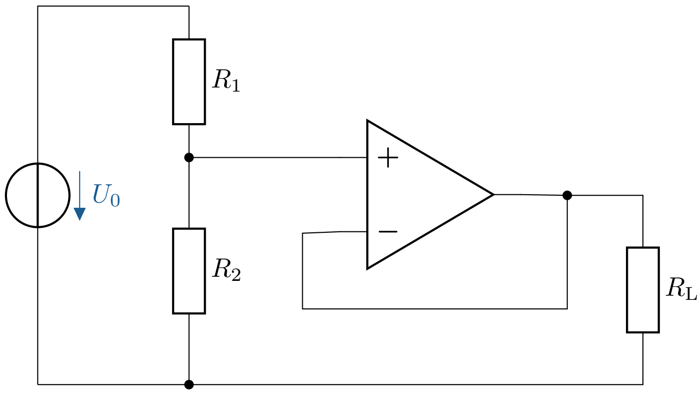
b) Calculation of the output voltage U_C with the connected load resistance R_L:

$$U_C = \frac{R_2}{R_1 + R_2 + R_L} \cdot U_{CC}$$
$$U_C = \frac{10\,\text{k}\Omega}{10\,\text{k}\Omega + 10\,\text{k}\Omega + 2{,}2\,\text{k}\Omega} \cdot 15\,\text{V}$$
$$U_C = \frac{10\,\text{k}\Omega}{22\,\text{k}\Omega} \cdot 15\,\text{V}$$
$$U_C = \frac{10}{22} \cdot 15\,\text{V}$$
$$U_C = \frac{150}{22}\,\text{V}$$
$$U_C \approx 6{,}82\,\text{V}$$

c) The voltage U_C decreases due to the load resistor R_L, as it forms a voltage divider with resistors R_1 and R_2. The load resistor R_L draws part of the voltage from the circuit, causing the output voltage U_C to drop.

See section: 3.2 and Table 3.1

B.20 Voltage divider with operational amplifier as voltage follower



a) Calculation of the output voltage U_C with and without load resistor
Since the operational amplifier is connected as a voltage follower (impedance buffer), the output voltage U_A exactly corresponds to the voltage across the voltage divider:

$$U_A = U_C = 7{,}5\,\mathrm{V}$$

Since the operational amplifier has a very high input resistance, the voltage is not affected by the load resistance. Thus, the output voltage remains constant at 7.5 V, regardless of whether R_L is connected or not.

b) Compare with task 13.
In contrast to the first circuit, the voltage does not drop when the load resistor R_L is connected. The operational amplifier decouples the voltage divider from the load, so that the voltage divider always maintains its original voltage.

c) Explanation of voltage stability through the operational amplifier
As an impedance converter, the operational amplifier has a high input resistance and a low output resistance. As a result, the load resistance is no longer directly connected to the voltage divider, but receives its current from the OPV. This means that the output voltage remains stable, regardless of the load.

See section: 3.2 and Table 3.1

B.21 Investigation of a step-up converter for voltage boosting

a) Explanation of how the step-up converter works

The step-up converter operates in two phases:

- **Phase 1 (switch closed):** The inductance L is charged with current and stores energy in its magnetic field. The diode blocks the current, so that no current flows to the output.
- **Phase 2 (switch open):** The stored magnetic flux of the inductance collapses, causing the inductance to induce a voltage. This voltage adds to the input voltage and charges the capacitor C to a higher voltage. The output now receives a higher voltage than the input voltage.

b) Derivation of the output voltage formula

In a steady state, the ratio between input and output voltage applies:

$$U_A = \frac{U_E}{1 - D}$$

where D is the duty cycle (ratio of the time during which the switch is closed).

c) Calculation of the output voltage for different D

Given is $U_\mathrm{E} = 1\,\mathrm{V}$.

- For $D = 0{,}3$:
$$U_\mathrm{A} = \frac{1\,\mathrm{V}}{1 - 0{,}3} = \frac{1\,\mathrm{V}}{0{,}7} = 1{,}43\,\mathrm{V}$$
- For $D = 0{,}5$:
$$U_\mathrm{A} = \frac{1\,\mathrm{V}}{1 - 0{,}5} = \frac{1\,\mathrm{V}}{0{,}5} = 2\,\mathrm{V}$$
- For $D = 0{,}7$:
$$U_\mathrm{A} = \frac{1\,\mathrm{V}}{1 - 0{,}7} = \frac{1\,\mathrm{V}}{0{,}3} = 3{,}33\,\mathrm{V}$$

d) Limits for $D \to 1$

Theoretically, $U_\mathrm{A} \to \infty$ if D approaches 1. In practice, however, there are limitations due to:

- Inductance saturation
- Losses due to resistors, diodes and switches
- Limited switching frequencies

Typical values for D are usually below 0.8 to ensure stable operation.